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Serhii S. Korniienko

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DISSERTATION
DIGITAL HISTORY TEACHING METHODOLOGY FOR FUTURE
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Dissertation contains the results of own research. All ideas, results, and texts of other authors have references to the relevant sources.

S. S. Korniienko

Scientific supervisor **Serhiy O. Semerikov**, Doctor of Educational Sciences, Professor,
Senior Researcher

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ABSTRACT

Корнієнко С.С. Методика навчання цифрової історії майбутніх фахівців. – Кваліфікаційна наукова праця на правах рукопису.

Дисертація на здобуття наукового ступеня доктора філософії за спеціальністю 011 Освітні, педагогічні науки. – Криворізький державний педагогічний університет, Кривий Ріг, 2026.

Дисертаційну роботу присвячено теоретичному обґрунтуванню, розробці та експериментальній перевірці методики навчання цифрової історії майбутніх фахівців. Дослідження спрямоване на вирішення наукової проблеми розробки методики навчання цифрової історії в закладах вищої освіти, яка забезпечує формування цифрових компетентностей та навичок критичного мислення.

У першому розділі подано теоретичні засади дослідження. За результатами систематичного огляду за методологією PRISMA було ідентифіковано 6 педагогічних досліджень, присвячених цифровій історії в освіті, що свідчить про початковий стан розробленості цієї проблеми. Систематичний огляд 74 досліджень гейміфікації в навчанні історії виявив, що лише 14,9% із них пропонують чітке визначення гейміфікації, 83,8% мають високий ризик упередженості за методологією MRQS, а 85,1% не надають операціональних визначень ключових конструктів. Аналіз якості досліджень за допомогою MRQS з валідацією великими мовними моделями (LLM) підтвердив ці результати та продемонстрував потенціал LLM для автоматизації оцінки якості досліджень. Розроблено рекомендації щодо впровадження гейміфікації на основі рамки цифрових компетентностей DigComp 2.0.

У другому розділі представлено методологію педагогічного експерименту. Дослідження проводилось на базі Центру довузівської підготовки та освіти Криворізького державного педагогічного університету протягом жовтня 2024 – кві-

тня 2025 року. У курсі підготовки до НМТ з історії України взяли участь 23 слухачі. Навчальний процес включав 38 лекцій, 30 семінарів із 156 тестовими формами Google Forms, 5 модульних контрольних робіт та інтерактивну гейміфіковану платформу HistoryLikeGame з 23 ігровими модулями. Збір даних здійснювався за допомогою Google Forms, журналу чату Viber та аналітики гейміфікованої платформи.

У третьому розділі подано результати педагогічного експерименту та їх обговорення. Аналіз відповідей студентів на семінарах 25–40 продемонстрував діапазон середніх результатів від 33,3% до 83,3%. Відслідковано індивідуальні траєкторії навчання 16 учасників. Аналіз відвідуваності виявив 33 повідомлення про відсутність, основними причинами яких стали проблеми з інтернетом (24%), сімейні обставини (24%) та відключення електроенергії (9%) – типові для умов воєнного часу. Гейміфікована платформа, впроваджена 1 грудня 2024 року, доповнила традиційні методи оцінювання інтерактивними елементами drag-and-drop та текстового введення.

Наукова новизна отриманих результатів полягає в тому, що: *вперше* розроблено та теоретично обґрунтовано методику навчання цифрової історії майбутніх фахівців, яка інтегрує гейміфікацію як ключовий компонент цифрової педагогіки; проведено систематичний огляд досліджень гейміфікації в навчанні історії за методологією PRISMA (74 дослідження) та оцінку їхньої якості за MRQS; здійснено валідацію автоматизованої оцінки якості досліджень за допомогою великих мовних моделей; *удосконалено* зміст та методи навчання історії з використанням цифрових засобів у закладах вищої освіти; *набули подальшого розвитку* теоретичні засади застосування гейміфікації та цифрових технологій у навчанні гуманітарних дисциплін.

Практичне значення отриманих результатів полягає в тому, що розроблено інтерактивну гейміфіковану платформу HistoryLikeGame та комплект навчально-методичних матеріалів для підготовчих курсів з історії України.

Ключові слова: цифрова історія, гейміфікація, методика навчання історії, систематичний огляд, PRISMA, MRQS, цифрова компетентність, DigComp 2.0, підготовка до НМТ, дистанційне навчання, освіта в умовах воєнного часу.

ABSTRACT

Korniienko S.S. Digital history teaching methodology for future professionals. – Qualifying scientific work on manuscript rights.

Dissertation for the Doctor of Philosophy degree in the speciality 011 Education and Pedagogical Sciences. – Kryvyi Rih State Pedagogical University, Kryvyi Rih, 2026.

This dissertation is devoted to the theoretical substantiation, development, and experimental verification of a digital history teaching methodology for future professionals. The research addresses the scientific problem of developing a methodology for teaching digital history in higher education institutions that ensures the formation of digital competencies and critical thinking skills.

Chapter 1 presents the theoretical foundations. A PRISMA-guided systematic review identified only 6 pedagogical studies dedicated to digital history in education, indicating the nascent state of the field. A systematic review of 74 studies on gamification in history education revealed that only 14.9% provided a clear definition of gamification, 83.8% carried a high risk of bias according to the MRQS methodology, and 85.1% failed to provide operational definitions of key constructs. A quality assessment using MRQS with large language model (LLM) validation confirmed these findings and demonstrated the potential of LLMs for automating research quality evaluation. Guidelines for implementing gamification based on the DigComp 2.0 digital competence framework were

developed.

Chapter 2 describes the experimental methodology. The study was conducted at the Centre for Pre-University Training and Education of Kryvyi Rih State Pedagogical University during October 2024 – April 2025. Twenty-three students participated in an NMT (National Multi-Subject Test) preparatory course in Ukrainian history. The educational process comprised 38 lectures, 30 seminars with 156 Google Forms quizzes, 5 modular controls, and an interactive gamified platform HistoryLikeGame with 23 game modules. Data were collected via Google Forms, a Viber chat log, and gamification platform analytics.

Chapter 3 presents the results and their discussion. Analysis of student responses across Seminars 25–40 demonstrated mean scores ranging from 33.3% to 83.3%. Individual learning trajectories were tracked for 16 participants. Attendance analysis revealed 33 absence reports, with primary reasons including internet issues (24%), family circumstances (24%), and power outages (9%) – characteristic of wartime conditions. The gamified platform, introduced on December 1, 2024, supplemented traditional assessment methods with interactive drag-and-drop and text-input elements.

The scientific novelty of the obtained results lies in the fact that: *for the first time*, a methodology for teaching digital history to future professionals that integrates gamification as a key component of digital pedagogy has been developed and theoretically substantiated; a PRISMA-guided systematic review of gamification studies in history education (74 studies) and their quality assessment using MRQS have been conducted; validation of automated research quality assessment using large language models has been performed; *improved*: the content and methods of history education using digital tools in higher education institutions; *further developed*: the theoretical foundations for applying gamification and digital technologies in humanities education.

The practical significance of the obtained results lies in the development of the interactive gamified platform HistoryLikeGame and a set of educational and methodological materials for Ukrainian history preparatory courses.

Keywords: digital history, gamification, history teaching methodology, systematic review, PRISMA, MRQS, digital competence, DigComp 2.0, NMT preparation, distance learning, wartime education.

List of education seeker publications

Scientific works in which the main scientific results of the dissertation are published

1. Korniienko, S. S. (2025). Digital history in educational research: A PRISMA-guided systematic review of pedagogical studies. *CTE Workshop Proceedings*, 12, 236–258. **(article in a foreign scientific publication indexed by Scopus)**
2. Korniienko, S. S., & Semerikov, S. O. (2026). Gamification in history education: A systematic review and MRQS quality assessment with LLM validation. *Educational Technology Quarterly*. **(article in a journal included in the list of specialized scientific publications of Ukraine, indexed by Scopus)**
3. Korniienko, S. S. (2025). Gamification in history education: A systematic review of definitions, effectiveness, and design principles. *Educational Dimension*. **(article in a journal included in the list of specialized scientific publications of Ukraine)**
4. Titova, N. M., Korniienko, S. S., Zahorodko, P. V., Moiseienko, M. V., & Donchev, I. D. (2025). Gamification of learning history through the lens of digital competence: Implementation guidelines based on DigComp 2.0. *CTE Workshop Proceedings*. **(article in a foreign scientific publication indexed by Scopus)**

Scientific works certifying the approbation of dissertation materials

5. Korniienko, S. S. (2025). Digital tools in history education: A systematic approach.

In *Proceedings of the 12th Workshop on Cloud Technologies in Education (CTE 2025)*, Kryvyi Rih, Ukraine.

6. Korniienko, S. S., & Semerikov, S. O. (2025). Methodological quality of gamification research in history education. In *Proceedings of the 5th International Conference on History, Theory and Methodology of Learning (ICHTML 2025)*, Kryvyi Rih, Ukraine.
7. Korniienko, S. S. (2026). Gamification-enhanced history education under wartime conditions. In *Proceedings of the Educational Technology Quarterly Workshop (ETQ 2026)*, Kryvyi Rih, Ukraine.

Scientific works additionally reflecting the scientific results of the dissertation

8. Korniienko, S. S. (2024). HistoryLikeGame: An interactive gamification platform for Ukrainian history education. Software registration certificate.

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INTRODUCTION

Justification of the choice of the research topic. In the National report on the state and prospects of the development of education in Ukraine, informatisation, digitalisation, and intellectualisation of all spheres of life are indicated as one of the basic transformational processes, and the digitalisation of education is a determining factor in its development [78]. Digital history in science is an interdisciplinary approach to the study of history by digital means [35]. In education and teaching methods, digital history acts as a new approach to the organisation of educational material and the learning process: creating interactive maps, processing large datasets, recognising handwritten texts, and creating infographics all facilitate mastery of educational content, thereby expanding students' cognitive abilities [14, 230, 242].

Elements of digital history are already used in the educational system of Ukraine at all levels. Since 2019, with the onset of the pandemic and later the full-scale Russian invasion, there has been an active move towards online courses and distance education, as well as localised use of various digital tools. In this context, gamification – the application of game design elements to non-game contexts [54] – has emerged as the most extensively researched digital approach to history education: our systematic review of 74 studies [124] confirmed that gamification dominates the literature on digital tools in history pedagogy. The encyclopedia definition frames gamification as the use of game elements and game-thinking in non-game activities [32]. Consequently, this dissertation positions gamification as a key subset of the broader digital history methodology, consistent with the UDC code [004.9+930]::378.147.

Digital history and its gamification component address several interconnected challenges:

1. Formation and development of digital competencies of educational process partic-

- ipants, aligned with the European DigComp 2.0 framework [232, 238];
2. Improvement of the learning process through interactive, evidence-based assessment tools [188, 198];
 3. Strengthening ties with the European educational research community and advancing integration processes.

Even though the use of Information and Communication Technologies (ICT) in education is thoroughly represented in the studies of the last decade [56, 122], the methods of ICT application cannot be directly transferred to history education without adaptation. In the framework of globalisation and informatisation, there is a need for a new, systemic and specialised approach. Digital history allows combining existing methodological assets with new methods, techniques, and tools that exist at the intersection of education, science, and technology [35, 242]. At the same time, for use in both higher and general secondary education, this approach needs experimental verification and scientific substantiation [92, 225], which should first be realised with future specialists as more mobile and independent learners.

Thus, the relevance of the scientific understanding of the research problem is due to contradictions between:

- modern needs of society and the insufficient degree of implementation of digitalisation and intellectualisation in history education [78];
- the importance of the formation of sociocultural competence of future specialists in the conditions of digitalisation and the absence of coherent theoretical and experimental justification of this process [56, 89];
- the objective need for scientific-methodical support of the process of formation of information competence in future specialists and the insufficient development of the methodology of teaching digital history [14];

- the growing body of gamification research (74 studies identified) and the critically low methodological quality of existing studies (83.8% with high risk of bias per MRQS assessment) [125].

Connection of work with scientific programs, plans, topics. The dissertation work was carried out in accordance with the plans of the scientific research work of the Department of Pedagogy of Kryvyi Rih State Pedagogical University and the topic of the complex scientific study “Methodology of teaching Digital History to future specialists” (State registration number 0120U101116 dated 26.02.2020).

The purpose of the study is to theoretically substantiate, develop, and experimentally test the methodology of teaching digital history for future professionals.

The object of research is the educational process of training future professionals in digital history methodologies within higher education institutions.

The subject of the study is the pedagogical methodology for teaching digital history to future professionals, including its theoretical foundations, practical implementation strategies, and effectiveness assessment.

Primary research question. How can the teaching methodologies of digital history be designed to effectively prepare students for professional careers in digital historical fields while maintaining academic rigour and the excellence of historical thinking?

Secondary research questions:

1. How do current theoretical frameworks in digital humanities pedagogy address the specific needs of professional training in digital history?
2. What methodological quality characterises existing research on gamification in history education, and what gaps need to be addressed? [125]
3. What institutional, technological, and wartime conditions affect equitable access to digital history tools in professional training environments?

4. What design principles for gamification are most effective in the context of history education? [54, 124]

Research hypothesis. The implementation of a gamification-enhanced digital history teaching methodology will improve student engagement and learning outcomes in history education [89, 198], as measured by repeated assessment performance, while the systematic analysis of existing research quality will provide evidence-based guidelines for future implementations [125, 177].

Research methods. To achieve the stated purpose, the following methods were employed:

- *Theoretical methods:* analysis and synthesis of scientific literature; systematic review following the PRISMA 2020 guidelines [177]; methodological quality assessment using the Mixed Methods Research Quality Synthesis (MRQS) tool; comparative analysis of gamification definitions and frameworks [54, 56].
- *Empirical methods:* pedagogical experiment (ex post facto single-group repeated-measures design); quantitative analysis of Google Forms quiz responses; qualitative content analysis of the course chat log; analysis of gamification platform usage.
- *Statistical methods:* descriptive statistics (mean, median, standard deviation); score trajectory analysis; attendance pattern analysis.
- *Validation methods:* LLM-assisted quality assessment validation; cross-referencing of automated and manual MRQS scoring [125, 131].

Scientific novelty of the obtained results.

For the first time:

- a comprehensive methodology for teaching digital history to future professionals that integrates gamification [54] as a key component of digital pedagogy has been

developed and theoretically substantiated;

- a PRISMA-guided systematic review of pedagogical studies on digital history in education has been conducted, identifying 6 relevant studies and establishing the nascent state of the field [123];
- a systematic review of 74 studies on gamification in history education has been performed, revealing that only 14.9% provide clear definitions of gamification and 83.8% exhibit high risk of bias [124];
- the methodological quality of gamification research in history education has been assessed using MRQS with LLM validation, demonstrating the potential of automated quality assessment [125];
- an experimental study of gamification-enhanced history education under wartime conditions has been conducted, documenting the unique challenges of distance learning during armed conflict [127].

Improved:

- the content and methods of history teaching using digital tools in higher education, through the development of an integrated assessment system combining 156 Google Forms quizzes, 7 modular controls, and an interactive gamified platform;
- guidelines for implementing gamification in history education based on the Dig-Comp 2.0 digital competence framework [232].

Further developed:

- the theoretical foundations for applying gamification and digital technologies in humanities education [124, 127];
- the methodology for systematic reviews in educational research, particularly regarding quality assessment with MRQS and LLM validation tools [125].

Personal contribution of the education seeker. All scientific results presented

in this dissertation were obtained personally by the author. In co-authored publications:

- in the paper co-authored with S. O. Semerikov on MRQS quality assessment with LLM validation [125], the author personally conducted the systematic review, performed the MRQS assessment, designed and executed the LLM validation experiment, and wrote the manuscript; the co-author provided supervision and methodological guidance;
- in the paper co-authored with N. M. Titova, P. V. Zahorodko, M. V. Moiseienko, and I. D. Donchev on gamification and DigComp 2.0 [232], the author personally developed the gamification platform, designed the implementation guidelines, and contributed to the theoretical framework and manuscript preparation.

Approbation of dissertation materials. The main provisions and results of the dissertation were presented and discussed at:

- the 12th Workshop on Cloud Technologies in Education (CTE 2025), Kryvyi Rih, Ukraine, 2025;
- the 5th International Conference on History, Theory and Methodology of Learning (ICHTML 2025), Kryvyi Rih, Ukraine, 2025;
- the Educational Technology Quarterly Workshop (ETQ 2026), Kryvyi Rih, Ukraine, 2026.

Publications. The main results of the dissertation are published in 8 scientific works [123–125, 127, 232], including: 4 articles (2 in publications indexed by Scopus, 1 in a journal included in the list of specialised scientific publications of Ukraine, 1 submitted to a specialised journal); 3 conference papers; 1 software registration certificate.

Structure and volume of the dissertation. The dissertation consists of an introduction, three chapters, conclusions, a list of references, and appendices. The total volume of the dissertation is 195 pages, of which 175 pages constitute the main text. The

dissertation contains 18 tables, 13 figures. The list of references includes 294 items.

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CHAPTER 1

THEORETICAL FOUNDATIONS OF TRAINING FUTURE SPECIALISTS USING DIGITAL HISTORY

1.1. Digital history in pedagogical research

Digital history is a modern interdisciplinary approach to the study of history using digital means [83, 156, 195]. There are various interpretations of this term, but they all relate to society, research, teaching, and communication. Digital history involves creating digital archives, databases, and multimedia resources to make historical data more accessible and interactive [35]. Unlike traditional historical methodology, digital history does not offer a revolutionary new approach but provides relevant tools in the era of information technologies, enabling the performance of scholarly work in significantly less time [36, 81, 82].

The concept of “digital history” emerged in the mid-1990s alongside the development of the World Wide Web [187, 207]. Roy Rosenzweig and Daniel Cohen at the Center for History and New Media (George Mason University) were among the pioneers who articulated the theoretical foundations of this field [36]. Since then, digital history has expanded to encompass a broad range of activities including text mining, Geographic Information Systems (GIS), digital storytelling, virtual museum exhibits, and computational approaches to historical analysis [17, 48, 49, 82, 142].

1.1.1. Methodology of the systematic review

To establish the current state of digital history in pedagogical research, we conducted a systematic review following the PRISMA 2020 guidelines [177]. The review aimed to answer the primary research question [124]: *How effective is teaching digital*

history to students compared to non-digital methods of teaching history in terms of improving learning outcomes?

Subsidiary research questions addressed [124]: (1) the effect of course status (mandatory vs. elective) on effectiveness; (2) which specific digital tools demonstrate the highest effectiveness; (3) which outcome indicators are most sensitive to digital methods; and (4) whether intervention duration affects effectiveness.

Search strategy. The Scopus database was selected as the primary information source due to its comprehensive coverage of peer-reviewed literature across disciplines [61], indexing over 48,000 active titles from more than 5,000 international publishers. The search string employed Boolean operators and was applied to title, abstract, and keyword fields:

TITLE-ABS-KEY ("digital history" AND student*)

The query was further limited to English-language publications and to document types including journal articles, conference papers, and book chapters. The search was executed on July 26, 2025, with no date restrictions applied during the search phase.

Inclusion criteria. Studies were included if they: (1) involved undergraduate or master's-level students enrolled in history courses at accredited higher education institutions, aged 18+; (2) implemented any digital technology intervention designed to enhance history education (collaborative writing platforms, digital archives, GIS, virtual reality, gamification, digital storytelling, computational text analysis) [6, 14, 29, 44, 51, 91, 114, 121, 169, 230, 242]; (3) employed quantitative, qualitative, or mixed methods with empirical data; (4) were published as peer-reviewed articles, conference proceedings, or book chapters between 2011 and 2025.

Exclusion criteria. Studies were excluded if they: (1) involved pre-university (K-12) populations exclusively; (2) were theoretical or conceptual papers lacking empir-

ical data [15, 170, 243]; (3) focused on technology training for educators without student outcome assessment; (4) addressed disciplines other than history; (5) involved interventions lasting less than two weeks or fewer than five participants; (6) were published in languages other than English.

Selection process. The selection process followed a two-stage screening protocol [124]. The initial search yielded 46 records from Scopus. After automated deduplication ($n=1$), 45 unique citations were screened by title and abstract. To support the screening process, Claude 4 Opus and Claude 4 Sonnet were used as auxiliary tools for preliminary analysis, with all final inclusion/exclusion decisions made by human reviewers [148, 226]. At the title-abstract screening stage, 28 records were excluded: 11 involved pre-university populations, 4 examined unrelated interventions, and 13 lacked empirical data.

Of the 17 studies advancing to full-text assessment, 2 could not be retrieved despite attempts to contact authors. The remaining 15 studies were assessed against all inclusion criteria using both human and LLM assistance. Nine additional studies were excluded: 2 were published outside the eligibility timeframe, 3 involved graduate students exclusively, 2 presented insufficient methodological detail, and 2 described theoretical frameworks without empirical testing. Ultimately, 6 studies satisfied all inclusion criteria.

Data collection. Data from each included study were collected by three reviewers simultaneously: two generative AI-based services (Claude 4 Sonnet and Grok 3) and one human researcher [148, 197]. The LLMs worked with identical prompts to ensure standardisation. When discrepancies were found, responses from all three reviewers were compared, with final decisions made by the human researcher. A standardised data extraction form captured study characteristics, participant demographics, intervention

details, outcome measures, and quality indicators [96, 118].

1.1.2. Results of the systematic review

The identification of only 6 studies meeting the inclusion criteria underscores the nascent state of digital history as a pedagogical research area [124]. This finding is particularly striking given the widespread use of digital tools in historical research [48, 82] and the growing emphasis on digital competencies in education policy [236, 238]. All six studies were published in English in peer-reviewed venues between 2011 and 2017 [12, 39, 52, 133, 153, 213].

1.1.2.1. Characteristics of included studies

Table 1.1 presents the key characteristics of the six included studies, and Table 1.2 summarises the intervention details.

Table 1.1: Characteristics of included studies in the digital history systematic review

Authors	Design	Country	Participants	Level	Duration
Bell (2016)	Case study	Australia	~100	UG	1 semester
Coleborne (2011)	Case study	New Zealand	43	UG	3 weeks
Davis (2017)	Case study	USA	Not specified	UG	1 semester
Lee (2014)	Design-based	USA	200 (total)	UG	5 years
McLean (2014)	Mixed methods	Canada	27 (8 resp.)	MA	1 semester
Soh (2013)	Quasi-experimental	USA	150	UG	1 semester

The six studies represent diverse geographical contexts: three from the United States [52, 133, 213], and one each from Australia [12], New Zealand [39], and Canada [153]. Sample sizes ranged from 27 to approximately 200 participants, with the sole quasi-experimental study [213] having the second-largest sample (150 students). Only one study included a control group, reflecting the common challenge of implementing randomised designs in educational settings [203].

Table 1.2: Intervention details for included studies

Authors	Intervention	Technologies	Control	Outcomes
Bell (2016)	3-minute digital video stories	Various digital tools	None	Engagement, understanding
Coleborne (2011)	Digital storytelling	Windows Movie Maker	None	Storytelling skills, historical thinking
Davis (2017)	Digital exhibition	Omeka (web)	None	Motivation, digital skills
Lee (2014)	Personal digital histories	Various (iterative)	None	Engagement, technology skills
McLean (2014)	Historical projects	Wiki, blogs	None	Technology skills, collaboration
Soh (2013)	Collaborative writing	ClassroomWiki	Traditional	Writing skills, engagement, learning

1.1.2.2. Study findings

The included studies documented the following interventions and findings:

1. *Digital storytelling with video* [12]: Approximately 100 undergraduate students created 3-minute digital video stories about historical topics. The case study documented enhanced engagement with historical content and deeper understanding of historical concepts through the process of narrative construction. The digital format facilitated students' spontaneous pursuit of additional primary sources beyond course requirements.
2. *Digital storytelling with multimedia* [39]: Forty-three undergraduates used Windows Movie Maker to create digital stories about historical events. The study found improvements in both digital storytelling skills and historical thinking, demonstrating that multimedia production can serve as a vehicle for developing analytical capabilities.
3. *Digital exhibition creation* [52]: Students used the Omeka web platform to create public-facing digital exhibitions incorporating metadata standards and Dublin

Core elements. The study reported that 45% of students expressed enthusiasm about the digital format, and students demonstrated improved digital skills alongside historical learning.

4. *Personal digital histories* [133]: This design-based research study spanned 5 years with 8 iterative cycles, involving approximately 200 students. Students created personal digital histories using various technologies. The iterative refinement process revealed that student engagement was highest when digital tools were directly aligned with historical inquiry tasks [191].
5. *Wiki and blog-based projects* [153]: Twenty-seven master's students (8 of whom provided survey responses) used wiki and blog platforms for collaborative historical research projects. The mixed-methods study found improvements in technology skills and collaborative competencies, though the small response rate limited generalisability.
6. *Collaborative wiki writing* [213]: The only quasi-experimental study, involving 150 undergraduate students using ClassroomWiki compared to traditional approaches. The study documented a 340% increase in peer-to-peer interactions compared to traditional discussion formats, with students spending an average of 12.4 hours on digital assignments (approximately 40% more than on traditional assignments). Writing skills, engagement, and learning outcomes all showed positive effects.

1.1.3. Cross-cutting analysis of outcomes

Analysis of primary and secondary outcomes across the six studies revealed consistent patterns.

Primary outcomes. Historical knowledge acquisition was assessed in four studies, with three (67%) reporting positive effects. Soh, Khandaker and Thomas [213] provided the strongest evidence, with students in the wiki-based condition demonstrating

significantly better writing quality and content accuracy than the control group. Lee and Molebash [133] documented improved engagement with historical evidence across eight iterative cycles. Bell et al. [12] and Coleborne and Bliss [39] reported enhanced understanding of historical concepts through the digital storytelling process, though without quantitative comparison to control conditions.

Secondary outcomes. Student engagement was the most consistently positive outcome, with all six studies (100%) reporting positive effects on at least one dimension of engagement [12, 39, 52, 133, 153, 213]. Behavioural engagement was measured through platform analytics (time-on-task, number of contributions, peer interactions) and showed the most dramatic improvements. Cognitive engagement manifested through enhanced depth of historical analysis: Bell et al. [12] reported that students spontaneously pursued additional primary sources beyond course requirements. Source analysis skills improved in three studies [52, 133, 153], particularly those employing digital archive platforms. The development of digital skills emerged as an unexpected but valuable outcome, with students acquiring competencies in metadata creation, digital curation, and platform-specific technical skills [52, 153].

Mechanisms of effectiveness. Three mechanisms appeared to drive positive outcomes across studies [124]: (1) *multimodal representation*, enabling students to engage with history through diverse sensory channels [12, 39]; (2) the “*authentic audience*” effect, where knowing their digital products would be publicly visible motivated higher-quality work [52, 133]; and (3) *technological affordances* that enabled pedagogical approaches impractical in analog environments (version control making thinking visible, collaborative platforms distributing cognitive load) [153, 213].

Barriers and challenges. Despite positive outcomes, implementation challenges were documented [124]. Technical difficulties consumed substantial instructional time

(one study reported 15% of sessions devoted to troubleshooting) [153]. Instructor preparedness was a critical barrier, with faculty struggling to integrate technological and pedagogical knowledge effectively [52, 133]. Assessment challenges were also noted: traditional evaluation methods proved inadequate for capturing learning in digital environments [39].

1.1.4. Analysis and implications

The small number of identified studies reveals a significant gap between the practice of digital history in research and its systematic integration into pedagogy. Several factors may explain this gap:

First, digital history has developed primarily as a research methodology rather than a teaching approach [48, 82]. The pioneers of the field (Rosenzweig, Cohen) focused on demonstrating the utility of digital tools for historical investigation [36], with less attention to how these tools can be effectively taught to students. The six included studies all represent individual instructors' initiatives rather than systematic programme-level implementations.

Second, the interdisciplinary nature of digital history creates challenges for pedagogical research [17, 142, 143]. Studies must address competencies in both history and technology, requiring expertise that spans traditional disciplinary boundaries. The TPACK framework, referenced in four of the six studies [12, 52, 133, 153], provides a theoretical lens for understanding this intersection but does not resolve the practical challenge of designing interventions that develop both types of knowledge.

Third, the rapid evolution of digital tools means that pedagogical approaches may become obsolete before they can be rigorously evaluated. This creates a tension between innovation and evidence-based practice. The included studies span 2011–2017, using platforms (ClassroomWiki, Windows Movie Maker, Omeka) that have evolved

substantially since publication [251].

Fourth, the absence of standardised outcome measures for digital history pedagogy makes it difficult to compare results across studies or to accumulate evidence systematically [92, 225]. Unlike STEM education, where standardised tests provide common outcome metrics, history education relies on diverse assessment approaches (essays, projects, portfolios) that resist direct comparison [149, 161, 193, 206].

The scarcity of rigorous pedagogical research on digital history motivates the broader examination of gamification in history education presented in the following sections, as gamification represents the most extensively researched digital approach to history teaching [41, 56, 122]. While the digital history review identified only 6 studies, the gamification review [126] identified 74 – a 12-fold difference that underscores the disproportionate research attention that game-based approaches have received.

1.1.5. Conclusions to Section 1.1

The PRISMA-guided systematic review [124, 177] demonstrates that digital history in pedagogical research remains an underdeveloped field, with only 6 studies meeting rigorous inclusion criteria [12, 39, 52, 133, 153, 213]. This finding establishes the need for: (1) more empirical research on digital history pedagogy; (2) examination of specific digital approaches (such as gamification) that have been more extensively studied [41, 188]; and (3) development of comprehensive teaching methodologies that integrate multiple digital tools [236].

1.2. Gamification in history education: definitions, effectiveness, and design principles

1.2.1. Background and rationale

Gamification – the application of game design elements to non-game contexts [54] – has attracted significant research attention in education over the past decade [56, 89, 122, 198]. In history education specifically, gamification offers opportunities to enhance engagement, motivation, and learning through interactive elements such as points, badges, leaderboards, quests, and narrative structures [41, 184, 188].

Given the paucity of research on digital history pedagogy identified in Section 1.1, this section presents a comprehensive systematic review of gamification in history education [126] as the most extensively researched component of digital history methodology.

1.2.2. Methodology

The systematic review followed the PRISMA 2020 guidelines [177]. Four research questions guided the review:

1. How is gamification defined and conceptualised in history education?
2. What is the current state of research on gamification effectiveness in history education?
3. What are the key design principles, strategies, and technologies used?
4. What are the main challenges, limitations, and potential negative effects?

Search strategy. Three databases were searched: Scopus [61], Web of Science [33], and Dimensions [57]. The search strategy combined gamification-related terms with history education terms in paper titles:

TITLE (history AND (game* OR gamification) AND (education OR teaching OR learning OR student* OR instruction))

Limits were applied for time frame (2000–2024), language (English or Ukrainian),

and publication type (peer-reviewed articles, conference papers, book chapters).

Inclusion criteria: (1) students in primary, secondary, or higher education learning history; (2) gamification techniques, elements, or strategies in history education; (3) empirical studies using quantitative, qualitative, or mixed methods; (4) formal or informal educational settings; (5) published between 2000 and 2024; (6) indexed in Scopus, Web of Science, or Dimensions.

Screening process. The initial search yielded 4,011 records across the three databases [126]. After duplicate removal, title-abstract screening, and full-text assessment, 80 studies advanced to data extraction. Of these, 6 were excluded during detailed analysis (3 were not about history education, 1 was a systematic review, 1 was about the history of mathematics, and 1 contained no gamification intervention), yielding 74 studies published between 2004 and 2024 across 27 countries.

Data extraction was automated using GPT-4o with a structured 21-item prompt, followed by human verification [148, 197]. Risk-of-bias assessment employed an adapted ROBIS tool with three domains and six signalling questions [96, 241].

1.2.3. Definitional analysis

1.2.3.1. Prevalence of definitional ambiguity

A critical finding of this review concerns the use of definitions. Of the 74 included studies:

- Only 11 studies (14.9%) provided a clear, explicit definition of gamification;
- 63 studies (85.1%) used the term without providing an operational definition;
- Among those providing definitions, the most commonly cited was that of Deterding et al. (2011): “the use of game design elements in non-game contexts” [54];
- Several studies conflated gamification with game-based learning, serious games, or educational games without distinguishing between these related but distinct

concepts [106, 134, 201].

This definitional ambiguity has significant implications for research synthesis and practice [32, 56, 122]. Without clear definitions, it is difficult to compare findings across studies, replicate research, or develop evidence-based guidelines for practitioners [20, 34].

The definitional approaches differed systematically by disciplinary orientation. Studies rooted in educational technology and human–computer interaction tended to define gamification narrowly as the addition of game elements to non-game contexts, emphasising mechanics such as points, leaderboards, and badges [54, 205]. In contrast, studies from history education and the humanities conceptualised the approach more broadly as game-based or experiential learning, foregrounding narrative immersion, historical empathy, and perspective-taking [3, 77, 216]. A third group of studies from computer science and design framed the concept through a software engineering lens, emphasising system design and usability [215, 220]. This disciplinary fragmentation helps explain both the terminological inconsistency and the diversity of theoretical frameworks employed [20, 23].

Table 1.3 presents the 11 studies that provided explicit definitions. These definitions span a range of conceptualisations, from broad frameworks for game-based learning to specific definitions of gamification, serious games, pervasive games, alternate reality games, and simulation games. The diversity in terminology and scope underscores the field’s lack of definitional consensus.

1.2.3.2. Conceptual approaches

We identified four distinct conceptual approaches to gamification in the literature:

1. **Structural gamification:** Adding game elements (points, badges, leaderboards) to existing educational content without modifying the content itself [54, 205].

Table 1.3: Studies providing explicit definitions of gamification or related terms (n=11 of 74)

Study	Term defined	de-	Definition (excerpt)
Mz and Sy [165]	Game-based learning		"A paradigm which utilizes the game as a medium for conveying the learning contents; leveraging the power of computer games to captivate and engage end users"
Zin, Yue and Jaafar [252]	DGBL		"A paradigm which utilizes the game as a medium for conveying the learning contents...for a specific purpose"
Corbeil and Laveault [43]	Simulation game		"Places [the participant] in a context, with certain givens, where he must attempt to achieve a goal"
Hadi, Wan Daud and Ibrahim [88]	Educational game		"A melding of educational content, learning principles, and computer games..."
Huffer and Oxenham [106]	Serious game		"Games that have an explicit and carefully thought-out educational purpose and are not intended to be played primarily for amusement"
Shih, Guo and Chen [208]	Pervasive game		"A mobile game that seamlessly integrates the physical environment with virtual scenarios"
Lynch, Mallon and Connolly [145]	Alternate reality game		"Cross media narrative-based games that use the Internet as a central communications platform"
Lestari, Basiroen and Ali [134]	Gamification		"A method of applying game mechanics to engage and motivate people more to complete their task voluntarily"
Scholz, Komornicka and Moore [201]	Gamification vs. GBL		"Gamification referring to the addition of game-based elements...game-based learning [as] more than isolated game elements"
Corrales Serrano [45]	Gamification		"A technique, a method and a strategy at the same time...starts from the knowledge of the elements that make games attractive"
Petersen et al. [180]	Serious game		"Applications that are not designed exclusively for fun or that are intended to be more than entertainment"

2. **Content gamification:** Transforming educational content to make it more game-like (narratives, characters, challenges, choices) [45, 134].
3. **Full gamification:** Comprehensive integration of game mechanics, dynamics, and aesthetics into the learning experience [198, 201].
4. **Game-based learning:** Using complete games (digital or analogue) as the primary pedagogical tool, which some studies classified under gamification [165, 184, 188, 252].

1.2.4. Characteristics of the evidence base

Before examining effectiveness findings, it is important to understand the characteristics of the 74 included studies, as these context factors influence the interpretation of results.

1.2.4.1. Geographic distribution

Research was conducted across 27 countries, but production was highly concentrated. Table 1.4 presents the complete distribution.

Table 1.4: Distribution of studies by country of application (n=74; some studies report multiple countries)

Country	n	Country	n
Taiwan	13	United States	11
Malaysia	9	Indonesia	8
Canada	6	Greece	5
Spain	4	United Kingdom	3
Sweden	3	Italy	3
Poland	3	Peru	2
Netherlands	2	Pakistan	1
Japan	1	Tunisia	1
Saudi Arabia	1	Chile	1
Portugal	1	Turkey	1
Denmark	1	Australia	1
China (Hong Kong)	1	Ireland	1
Austria	1	Germany	1
China	1		

Taiwan produced the most studies (n=13, 17.6%) [26, 27, 30, 102, 104, 137–139, 208–

210, 245, 246], followed by the United States (n=11, 14.9%) [64, 77, 80, 103, 111, 117, 196, 202, 212, 216, 219], Malaysia (n=9, 12.2%) [76, 88, 134, 144, 147, 159, 165, 247, 252], and Indonesia (n=8, 10.8%) [62, 65, 109, 112, 160, 167, 189, 190]. Together, these four countries accounted for 55.4% of the total. East and Southeast Asian countries (Taiwan, Malaysia, Indonesia, China/Hong Kong, Japan) contributed 43.2% of studies, reflecting strong regional interest in gamified history education. European countries contributed 33.8% of studies, with Greece (n=5) [7, 9, 116, 194, 227], Spain (n=4) [45, 73, 179, 224], and the United Kingdom, Sweden, Italy, and Poland (n=3 each) representing the most active contexts. The Global South (Latin America, Africa, Middle East, South Asia) was almost entirely absent, with only 6 studies from these regions combined. Notably, no studies from Ukraine or other post-Soviet states were identified, despite the region's rich historical traditions and growing digital education infrastructure [78].

1.2.4.2. Temporal distribution and research evolution

Studies spanned 2004–2024 with a pronounced increase from 2011 onward. Table 1.5 presents the yearly distribution.

Table 1.5: Yearly distribution of included studies (n=74)

Year	n	Year	n	Year	n	Year	n
2004	1	2010	2	2016	1	2022	4
2007	3	2011	7	2017	4	2023	7
2008	1	2012	1	2018	8	2024	3
2009	3	2013	2	2019	6		
		2014	7	2020	5		
		2015	5	2021	4		

The most productive years were 2011, 2014, 2018, and 2023 (each contributing 7–8 studies). The earlier period (2004–2017) and later period (2018–2024) each contributed 37 studies. Analysis of keyword evolution through bibliometric network analysis revealed three distinct temporal stages:

- 1. Early exploration stage (2004–2012):** Characterised by initial investigations

into using computer games for history education [7, 107, 165, 166, 202, 216]. Keywords such as *computer game*, *educational game*, and *learning* dominated this period. Research focused on foundational questions about whether games could effectively support history learning [181, 184].

2. **Narrative and engagement stage (2013–2017):** A shift toward examining how game narratives and storytelling can enhance historical understanding [38, 47, 77, 106, 117]. Keywords like *history learning*, *video game*, *motivation*, and *serious game* reflected growing interest in leveraging narrative elements for historical empathy and engagement.
3. **Immersive technology stage (2018–2024):** The adoption of emerging technologies [5, 28, 50, 53, 105, 129, 154], with keywords such as *augmented reality*, *virtual reality*, *gamification*, and *mobile game* indicating a broadening of technological approaches and a more systematic integration of game design principles into history education.

1.2.4.3. Publication types and database coverage

The majority of included studies were conference papers (n=40, 54.1%), followed by journal articles (n=24, 32.4%), book chapters (n=9, 12.2%), and one preprint (1.4%). Scopus was the most comprehensive source, covering 62 of 74 studies (83.8%). Dimensions contributed 11 unique studies (14.8%) not indexed in Scopus or Web of Science, demonstrating the value of searching multiple databases. The substantial proportion of conference papers suggests that much of the research in this area is preliminary, with findings presented at conferences before (or instead of) undergoing the more rigorous peer-review process of journal publication [126].

1.2.4.4. Education levels

Secondary education was the most studied context (n=35, 47.3%), followed by higher education (n=21, 28.4%) and primary education (n=13, 17.6%). Five studies (6.8%) addressed multiple levels or did not specify. The predominance of secondary education studies may reflect the alignment between game-based approaches and the developmental stage of adolescent learners, who are typically more receptive to gamified instruction [1, 107, 212, 228]. The relative scarcity of higher education studies is notable given that university students represent the target population for NMT preparatory courses [2].

1.2.4.5. Game types and technological formats

The interventions employed diverse formats, as shown in Table 1.6.

Table 1.6: Distribution of game types across included studies (n=74)

Game type	n	%
VR/AR games	19	25.7
Board and card games	11	14.9
Custom digital games	10	13.5
Web-based games	10	13.5
Commercial off-the-shelf	8	10.8
Other/mixed	7	9.5
Mobile games	5	6.8
RPG	3	4.1
Simulation	1	1.4

VR/AR games were the most common (25.7%), reflecting recent technological adoption [9, 50, 53, 71, 105]. Board and card games (14.9%) [43, 109, 129, 154, 218] and custom digital games (13.5%) [27, 30, 104, 137, 144] were also frequently used. Eight studies (10.8%) employed commercial off-the-shelf games such as *Civilization IV* [216] or *Assassin's Creed* [77]. Digital formats accounted for approximately 84% of the corpus.

The temporal evolution of game types reveals important trends. VR/AR technologies have been present since 2007 [7] but show markedly increased adoption from 2018 onward [28, 50, 105], coinciding with the decreased cost of VR headsets and the im-

proved accessibility of AR through smartphones. Custom digital games have been consistently popular throughout the period. Board and card games emerged as a notable category from 2014 [117, 129, 218], reflecting growing interest in non-digital gamification approaches that avoid technological barriers. Commercial off-the-shelf titles were sporadically used across the period, with notable examples including the Civilization series [80, 103, 111, 216], Assassin’s Creed [77, 196], and various historical strategy games [10, 70].

1.2.4.6. *Research designs*

The design landscape was dominated by non-experimental approaches. Table 1.7 presents the distribution of designs.

Table 1.7: Distribution of research designs across included studies (n=74)

Design	n	%
Qualitative	14	18.9
Quasi-experimental	13	17.6
Descriptive	10	13.5
Experimental (true)	9	12.2
Mixed methods	9	12.2
Design-based research	7	9.5
Pre-post (single group)	4	5.4
Case study	3	4.1
Survey	2	2.7
Other	3	4.1

The experimental family accounted for 36.5% (n=27), but only 1 study employed a fully randomised controlled design [203]. Quasi-experimental designs (n=13) [75, 107, 139, 246] and pre-experimental single-group pretest-posttest designs (n=9) [9, 30, 137, 140] made up the rest. Qualitative and case study designs constituted 21.6% (n=16) [77, 196, 216], design-based research 13.5% (n=10), and mixed methods 12.2% (n=9). Only 23 of 74 studies (31.1%) included a control or comparison group, and a substantial number (n=43, 58.1%) employed no quantitative comparative methodology at all. This methodological landscape stands in stark contrast to fields such as medical education or psy-

chology, where randomised controlled trials are the expected standard for effectiveness claims [86, 203, 217].

1.2.4.7. Sample sizes and statistical power

Among the 62 studies reporting participant numbers, the median was 33.5 (mean = 127.0, range: 4–2,417). The distribution was heavily right-skewed, as shown in Table 1.8.

Table 1.8: Distribution of sample sizes across included studies (n=74)

Sample size range	n	%
<30	25	33.8
30–99	23	31.1
100–499	9	12.2
≥500	5	6.8
Not reported	12	16.2

The prevalence of small samples (40.3% below 30, 66.1% below 50 among those reporting) has direct implications for statistical power [37]. A two-group comparison with medium effect size ($d = 0.50$) and $\alpha = .05$ requires approximately 64 participants per group (128 total) to achieve 80% power, a threshold exceeded by only a minority of studies. Studies with particularly small samples included Barbatsis et al. [9] (n=17), Nugroho et al. [167] (n=5), Ardito et al. [7] (n=4), and Froschauer et al. [71] (n=4). These small samples substantially limit generalisability and increase the likelihood of both Type I errors (due to inflated variance) and Type II errors (due to insufficient power). Furthermore, publication bias assessment through funnel plot analysis [175] was not feasible given the heterogeneity of outcome measures.

1.2.4.8. Theoretical frameworks

The studies cited diverse theoretical frameworks, with game-based learning frameworks being the most frequently referenced (27 studies) [165, 184, 188, 252], followed by flow theory (11 studies) [27, 30, 102], experiential learning and constructivism (9 studies each) [16, 42], collaborative learning and situated learning (8 studies each). Motivation

theories were cited in 7 studies [89, 198], as were historical thinking frameworks and design frameworks. The Technology Acceptance Model (TAM) appeared in 3 studies, and self-determination theory in 2.

Notably, discipline-specific frameworks such as historical thinking and historical empathy were cited in only 7 studies [3, 77, 196, 216], suggesting that gamification research in history education draws more heavily from general educational and psychological theories than from the epistemological traditions of the history discipline itself. This finding has implications for the theoretical grounding of gamification studies: without engagement with what it means to think historically [161, 193], gamification interventions risk developing surface-level knowledge (dates, names, events) rather than the deeper analytical skills that characterise expert historical reasoning.

1.2.4.9. Comparison with gamification in other domains

The characteristics of the history education evidence base can be contextualised by comparison with gamification research in other domains. A parallel systematic review of gamification in software engineering education [127] identified 29 studies from an initial pool of 486 records. Several patterns are consistent across domains:

- **Positive but uncertain effects:** Both reviews found predominantly positive effects on engagement and motivation, with more mixed evidence for learning outcomes [56, 122, 127]. In software engineering, the average effect on learning outcomes was small (Cohen's $d = 0.23$), with low certainty of evidence according to GRADE assessment [86, 87, 204].
- **Methodological limitations:** Both domains are characterised by small sample sizes, lack of control groups, and reliance on unvalidated instruments [34, 41]. In software engineering, 18 of 29 studies (62.1%) had serious overall risk of bias [127].
- **Game element preferences:** Software engineering gamification more commonly

employed structural elements (points in 17/29 studies, leaderboards in 11/29, badges in 9/29) [4, 11, 13, 55, 72, 90, 113, 152, 162, 183, 205, 215, 220], while history education relied more heavily on narrative and immersive elements (narrative/storytelling in 44/74 studies, immersion/VR/AR in 55/74).

- **Definitional consistency:** Neither domain showed strong definitional practices, though the history education review found even lower rates of explicit definition (14.9% vs. approximately 30% in software engineering) [127].

These cross-domain comparisons suggest that the methodological challenges identified in history education gamification research are not unique but rather reflect broader weaknesses in the gamification evidence base across educational domains [56, 89, 122, 163, 198].

1.2.5. Effectiveness findings

1.2.5.1. Outcome types measured

The review synthesised effectiveness evidence across multiple outcome categories [126]. The most frequently measured outcome was student engagement (52 studies, 70.3%), followed by knowledge acquisition (45 studies, 60.8%), attitudes and satisfaction (39 studies, 52.7%), and motivation (26 studies, 35.1%). Higher-order thinking skills, including historical thinking and empathy, were assessed in only 14 studies (18.9%) [3, 77, 117, 196, 216], indicating a gap in the evaluation of deeper cognitive and affective outcomes specific to the history discipline. Flow and immersion were measured in 15 studies [26, 27, 30, 102], collaboration in 18, and usability in 13.

1.2.5.2. Overall direction of effects

Of the 74 studies, 12 (16.2%) reported exclusively positive results across all measured outcomes, 37 (50.0%) reported mixed results (both positive and null or negative

findings across different outcome measures), 18 (24.3%) reported null or negative results, and 7 (9.5%) did not measure learning or engagement outcomes quantitatively [126]. The high proportion of mixed results underscores the complexity of gamification effects and the importance of examining specific outcome domains rather than drawing blanket conclusions about “effectiveness” [34, 56].

1.2.5.3. Student engagement and motivation

The majority of studies (n=52, 70.3%) reported positive effects of gamification on student engagement or motivation. Common findings included increased participation in voluntary activities, longer time-on-task, and higher self-reported motivation. Behavioural engagement metrics (platform analytics, participation rates) showed the most consistent positive effects. Specific quantitative findings included:

- A significant correlation between perceived reality and engagement ($r = .70$, $p < .001$) in a VR-based application [9];
- 73% of students positively motivated in a collaborative gaming study, with 93% having prior experience with gaming [239];
- 100% of surveyed students expressing willingness to play the game again in one mobile-based study [208];
- Mean engagement scores above 5 on a 7-point scale across multiple VR studies [53, 105];
- Over 90% of students showing positive motivation toward a game-based system, with satisfaction scores averaging 4.51 out of 5 [104];
- A significant gender difference in engagement, with boys demonstrating statistically higher engagement compared to girls ($F(1,64) = 4.408$, $p = .040$) [107].

However, the sustainability of these effects beyond initial novelty periods was rarely examined [56, 89]. Some studies documented that not all gamification strategies

were equally motivating: competitive timed games generated stress in some students [228], approximately 10% of participants in one study reported being negatively motivated [244], and female students were less familiar with or interested in strategy-based games [107]. These findings highlight the importance of considering individual differences and offering multiple pathways within gamified environments [198].

1.2.5.4. *Learning outcomes*

Results regarding academic achievement were more mixed [126]:

- 38 studies (51.4%) reported positive effects on learning outcomes;
- 28 studies (37.8%) reported no significant difference compared to control conditions;
- 8 studies (10.8%) reported negative or mixed effects [172, 244].

Table 1.9 presents a selection of studies reporting statistically significant quantitative learning outcomes. Among the studies reporting effect sizes, values ranged from small ($d = 0.28$) to medium ($d = 0.61$), with stronger effects observed for factual recall than for interpretive understanding.

Table 1.9: Selected studies reporting statistically significant learning outcomes

Study	n	Design	Outcome	Key statistic
Barbatsis et al. [9]	17	Pre-post	Knowledge	$Z = -2.226, p = .026$
Yu et al. [246]	103	Quasi-exp.	Achievement	$t = 2.01, p = .05$
Lin et al. [137]	27	Pre-post	Effectiveness	$t = 8.196, p < .001$
Chen, Hou and Chou [27]	38	Pre-post	Knowledge	$t = -5.33, p < .001$
Chou et al. [30]	66	Pre-post	Knowledge	$t = 9.58, p < .001$
Ghannem et al. [75]	36	Quasi-exp.	Knowledge	$W = 0, p \leq .01$
Lin and Wei [139]	30	Quasi-exp.	Achievement	Low achievers $p = .010$
Liu, Chan and Hou [140]	12	Pre-post	Knowledge	$Z = -2.77, p = .006$

This pattern suggests that gamification may be more effective for developing surface-

level knowledge than for fostering deep historical thinking [34, 41]. Qualitative studies provided additional evidence of learning enhancement, including increased correct responses in historical knowledge assessments [30, 75], higher retention scores in game-based learning compared to traditional methods (up to 9/10 vs. 3/10 correct answers) [134], increased student engagement and critical thinking [77, 196], improved team collaboration [218, 239], and the development of systemic understanding of history through recursive cycles of failure and strategy revision [111, 216].

1.2.5.5. Flow and immersion

Eight studies specifically measured flow states using validated instruments [26, 27, 30, 102, 104, 138, 210, 245], providing evidence that history games can achieve meaningful immersion. All reported flow scores above scale midpoints. Concentration and immersion consistently scored highest among flow dimensions, while loss of self-consciousness and time transformation scored lowest [27, 102]. One study found that flow state was significantly positively correlated with learning achievement ($p < .05$) [27], providing evidence that achieving flow states in history games may be a mechanism through which engagement translates into learning. Mean flow scores ranged from 3.63 to 4.23 on 5-point scales, with autotelic experience (enjoyment of the activity for its own sake) scoring particularly high ($M = 4.54$ in one study) [102].

1.2.5.6. Historical thinking skills

A smaller subset of studies ($n=18$) specifically examined the development of historical thinking skills such as source analysis, contextualisation, and perspective-taking [161, 193]. Of these, 12 reported positive outcomes, while 6 found no significant effects. Studies employing multi-perspectival digital archives appeared most effective, as students encountered conflicting accounts requiring critical evaluation [103, 117]. Narra-

tive video games were conceptualised as tools for fostering empathy and emotional connections to historical events and figures through role-play elements [3, 77, 196]. Board games were viewed as mechanisms for fostering historical perspective-taking through role-playing and strategic elements [43, 129, 218]. The limited number of studies assessing higher-order thinking skills (only 18.9% of the corpus) represents a significant gap, as these skills are central to the goals of history education [161, 193].

1.2.5.7. Affective outcomes and attitudes toward history

Studies examining affective outcomes such as attitude toward history, enjoyment, and self-efficacy generally reported positive results (n=31 of 37 reporting on these outcomes, 83.8%). The strongest positive effects were observed in studies using narrative-rich game environments, where historical content was embedded in engaging storylines [64, 77, 219]. One large-scale study (N = 2,417) identified five distinct engagement profiles through latent class analysis, finding that modifying game dialogue scripts shifted some students toward more positive evaluations of history learning [212]. Other studies documented transformations in students' relationships with history, with some students moving from viewing it as "boring" to developing genuine interest through game-based exploration [80, 216]. Structural equation modelling in one study confirmed that perceived usefulness had the strongest influence on game preference, and more sophisticated epistemological beliefs positively affected attitudes toward game-based history learning [245].

1.2.6. Implementation challenges

The review identified several categories of challenges that recurred across studies:

Technical constraints. Infrastructure requirements, software compatibility, and device availability posed persistent challenges [126]. VR/AR studies (25.7% of the corpus)

were particularly affected by hardware costs and maintenance demands [9, 50, 71]. Web-based and mobile formats showed lower technical barriers but introduced their own issues (browser compatibility, screen size limitations, connectivity requirements) [159, 208].

Resource limitations. Developing custom gamification environments required substantial investment in design, programming, and content creation [62, 74, 137, 144]. Studies using commercial off-the-shelf titles (10.8%) avoided development costs but sacrificed pedagogical control [25, 77, 216]. The resource intensity of gamification development may explain the prevalence of small-scale studies: few research teams have the resources to develop, deploy, and evaluate gamification at scale.

Curriculum integration difficulties. Fitting gamification into existing curricula and assessment structures was cited as a significant challenge [111, 115]. Time pressure was a common concern: gamified activities typically required more class time than traditional instruction, creating tensions with content coverage requirements [107, 218].

Teacher preparation. Instructor preparedness and confidence with gamification tools was identified as a critical factor [78, 188]. Studies where instructors received substantial training or co-designed the gamification elements reported better outcomes than those where technology was introduced without adequate pedagogical support [21, 115, 201].

Student resistance and equity. Some studies documented initial resistance from students who perceived gamification as trivialising serious historical content [172, 173, 244], or who were uncomfortable with competitive elements [99, 228]. Equity concerns were raised regarding differential access to technology and varying levels of digital literacy among students [66, 78, 236].

1.2.7. Game design elements and technological platforms

1.2.7.1. Game design element taxonomy

Analysis of the 74 studies identified a comprehensive taxonomy of game design elements utilised in history education gamification. Table 1.10 presents the frequency of major categories.

Table 1.10: Frequency of game design element categories across included studies (n=74)

Game design element category	n	%
Immersion/VR/AR	55	74.3
Narrative/Storytelling	44	59.5
Role-playing	44	59.5
Collaboration/Competition	36	48.6
Levels/Progression	35	47.3
Points/Rewards/Scoring	31	41.9
Exploration	29	39.2
Quiz/Assessment	19	25.7
Puzzles/Challenges	18	24.3
Rules/Goals	18	24.3
Feedback	17	23.0
Time pressure	12	16.2

Immersive technologies (VR/AR and 3D environments) were employed in 55 studies (74.3%), making them the most prevalent design element category [9, 28, 50, 53, 71, 105]. Role-playing and narrative/storytelling elements were each present in 44 studies (59.5%) [43, 64, 77, 209, 216], reflecting the natural alignment between historical content and narrative-driven game mechanics. Collaboration and competition mechanics appeared in 36 studies (48.6%) [107, 218, 227, 239], while points, rewards, and scoring systems – often considered the hallmark of gamification [54, 205] – were present in only 31 studies (41.9%). This finding suggests that history education gamification relies more heavily on narrative and experiential elements than on extrinsic reward systems, a pattern that distinguishes it from gamification in other domains (where points and leaderboards typically dominate) [127, 205, 215].

These elements can be further organised into functional clusters:

Exploration and navigation mechanics encompassed 3D exploration and interactive object manipulation [7, 9], virtual environment exploration [71, 105], location-based exploration and GPS-triggered events [40, 194, 208], open world exploration with narrative choice and branching [64, 216], first-person and third-person perspective navigation, and map-based navigation. These mechanics leveraged spatial cognition and discovery-based learning to engage students with historical environments.

Assessment and feedback systems included quizzes with increasing difficulty and immediate feedback [30, 45, 75], multiple-choice questions integrated into gameplay, progressive task difficulty and challenge systems [104, 137], performance tracking and progress monitoring, and real-time feedback through game mechanics [212, 228]. The integration of assessment within gameplay (rather than as separate evaluative events) was a distinguishing feature of effective implementations.

Narrative and storytelling elements comprised immersive storytelling and digital narratives [64, 77, 219], character-driven plots and NPC (non-player character) interactions [144, 209], historical scenario simulations and role-playing [43, 216], in-game cutscenes and cinematics, and storyline progression and chapter-based narratives [80, 196]. Historical narratives were adapted into game stories, real historical events served as plot bases, and historical figures were cast as interactive characters [38, 117].

Competition and collaboration mechanisms included team-based collaboration and cooperative gameplay [79, 107, 239], competitive elements and leaderboards [45, 134, 214], group competition and tournaments [58, 218, 227], and multiplayer functionality. The balance between competition and collaboration was identified as a critical design decision, as purely competitive structures risked discouraging lower-performing students [198, 228].

Reward and progression systems encompassed point-based scoring and re-

wards [104, 134], progression-based unlocking of content and level systems [30, 137], achievement records and badges [45, 54], and lives/health systems with resource management [27, 209].

Interactive historical content included historical artifact examination and collection [7, 117], interactive timelines and concept maps [157, 221], character cards and historical figure representations [109, 129], period-specific item cards [154, 218], and historical riddles and puzzles [73, 179].

Strategic and decision-making elements comprised strategy-based mechanics and tactical gameplay [80, 111, 216], resource management and economic systems [103], diplomatic relations and alliance systems, military conflict and combat systems [43], and historical decision-making scenarios [3, 244].

1.2.7.2. Technological platforms

The studies employed a diverse range of technological platforms [126]. Unity 3D/Unity Engine was the most frequently utilised game engine, appearing in 11 studies [9, 28, 50, 105], followed by Unreal Engine 4 (3 studies) [224]. Web-based technologies (HTML5, CSS3, JavaScript) were used in browser-based games [116, 159, 227], with PHP and MySQL for backend data management. VR/AR implementations ranged from low-cost solutions (Google Cardboard with smartphones) to high-fidelity systems (HTC Vive) [53, 71]. Commercial games repurposed for education included the Civilization series (6 studies) [80, 103, 111, 216], Assassin's Creed franchise (2 studies) [77, 196], and various complex historical strategy games (Crusader Kings II, Europa Universalis IV, Hearts of Iron IV) [135]. Non-digital platforms – physical board games, card-based games, and 3D printed models [43, 109, 129, 154, 218] – represented an important subset, demonstrating that gamification does not require digital technology.

Communication and collaboration platforms used alongside gamification included

Zoom for online learning, social media (Facebook) for community building, and learning management systems (Moodle) [19, 85]. This multi-platform approach foreshadows the multi-channel design adopted in the present study (Chapter 2).

1.2.7.3. Instructional design strategies

Collaborative learning was the most frequently employed instructional strategy (46 studies, 62.2%) [107, 218, 239], followed by narrative-driven instruction (40 studies, 54.1%) [64, 77, 219] and assessment-integrated approaches (31 studies, 41.9%) [30, 45, 75]. Scaffolded/guided approaches and problem/inquiry-based learning were each present in 23 studies (31.1%), as was experiential/situated learning [16, 42]. Constructivist approaches, while theoretically prominent in the literature, were explicitly employed in only 15 studies (20.3%), suggesting a gap between theoretical framing and actual instructional design [42].

Content integration approaches included immersive historical reconstruction (virtual reconstruction of archaeological sites) [7, 9, 53], narrative-driven learning (content delivered through storytelling) [64, 77], problem-based learning (historical challenges requiring application of knowledge) [106, 117], simulation-based learning (games modelling historical systems) [43, 216], role-playing and perspective-taking (students assuming roles of historical figures) [3, 209], artifact-based learning (interaction with digitised historical artifacts) [7, 157], location-based learning (GPS-triggered digital content at historical sites) [194, 208], and assessment-integrated learning (knowledge questions embedded as game challenges) [30, 75, 134].

1.2.8. Design principles

From the synthesis of effective interventions, six evidence-based design principles for gamification in history education were identified. These principles emerged from the

analysis of studies reporting positive outcomes and represent the common features of successful implementations.

1. **Historical authenticity:** The most effective implementations maintained historical accuracy while incorporating game elements [7, 9, 77]. Superficial gamification that trivialised historical content was associated with negative outcomes in 10.8% of studies [172, 244]. Authenticity encompasses both factual accuracy and appropriate representation of historical complexity, ambiguity, and multiple perspectives [161, 193]. Studies using historically accurate game environments (reconstructions of archaeological sites, period-appropriate visual design) reported stronger learning outcomes than those using generic game formats with historical content superimposed [9, 53, 157].
2. **Narrative integration:** Using historical narratives as the framework for gamified activities enhanced both engagement and learning [64, 77, 219]. Quest-based structures aligned naturally with historical investigation, casting students as detectives uncovering historical evidence [106, 117]. Narrative-rich environments showed the strongest effects on affective outcomes (attitude toward history, enjoyment) [212, 216]. Commercial titles like *Assassin's Creed* demonstrated the potential of narrative integration but raised concerns about historical liberties taken for entertainment purposes [77, 196].
3. **Progressive difficulty:** Scaffolded challenges that increased in complexity over time supported sustained engagement and skill development [30, 104, 137]. Effective scaffolding moved from recognition tasks (matching, multiple choice) to analysis tasks (source evaluation, perspective-taking) to synthesis tasks (constructing historical narratives, arguing causation). This progression mirrors established taxonomies of historical thinking skills [22, 161, 193] and ensures that gamification

supports, rather than undermines, cognitive development.

4. **Immediate feedback:** Providing instant feedback on quiz responses and game actions facilitated self-regulated learning [75, 212, 228]. Studies documented that immediate feedback enabled iterative learning cycles where students corrected misconceptions in real time [30, 45]. The feedback mechanism was particularly effective for factual recall tasks but less clearly effective for complex interpretive questions where multiple valid answers exist [34].
5. **Social elements:** Collaborative and competitive elements (team challenges, leaderboards, peer review) enhanced motivation, but their effects were moderated by implementation design [107, 198, 239]. Competition needed to be carefully balanced to avoid discouraging lower-performing students [228, 244]. Collaborative structures (team-based quests, cooperative problem-solving) showed more consistently positive effects than purely competitive elements [218, 227].
6. **Multiple modalities:** Combining text, images, maps, interactive elements, audio, and video addressed different learning preferences and supported deeper understanding [105, 157, 221]. The use of primary source images, historical maps, and audiovisual materials within gamified environments was associated with enhanced engagement across diverse learner profiles [64, 117]. VR/AR implementations provided the most immersive multimodal experiences but at higher cost and technical complexity [28, 50, 71, 101].

1.2.9. Conclusions to Section 1.2

The systematic review of 74 studies on gamification in history education [126], drawn from an initial pool of 4,011 records across three databases, reveals a rapidly growing but methodologically immature field. Key findings include:

- **Definitional ambiguity:** 85.1% of studies lack explicit definitions of gamification,

undermining construct validity and cross-study comparison;

- **Positive engagement effects:** 70.3% of studies report positive effects on engagement, but with limited evidence on long-term sustainability;
- **Mixed learning outcomes:** 51.4% report positive effects on learning, but effect sizes are small to medium ($d = 0.28\text{--}0.61$) and concentrated on factual recall rather than deep understanding;
- **Design diversity:** VR/AR (25.7%), board games (14.9%), custom digital (13.5%), and web-based (13.5%) formats dominate, with digital approaches accounting for 84% of the corpus;
- **Geographic concentration:** Four countries (Taiwan, USA, Malaysia, Indonesia) account for 55.4% of all studies;
- **Methodological weaknesses:** Only 31.1% include control groups, median sample size is 33.5, and only 1 study is an RCT.

The six design principles identified (historical authenticity, narrative integration, progressive difficulty, immediate feedback, social elements, multiple modalities) provide an evidence-based foundation for implementation, which informs the experimental design presented in Chapter 2.

1.3. Methodological quality of gamification research in history education

1.3.1. Rationale for quality assessment

The positive findings reported in Section 1.2 must be interpreted in light of the methodological quality of the underlying research [34, 56, 122]. Previous meta-analyses in educational technology have shown that studies with higher methodological quality tend to report smaller effect sizes, suggesting that methodological weaknesses may

inflate perceived effectiveness [92] [41, 225].

This section presents a quality assessment of the 74 studies identified in the systematic review [126], using the Mixed Methods Research Quality Synthesis (MRQS) tool [125] with an innovative LLM validation approach.

1.3.2. The MRQS framework

To address the absence of a structured quality assessment instrument adapted for gamification research in history education, we developed the Methodological Reporting Quality Score (MRQS) [125] – a ten-item binary instrument (scored 0 or 1 per item, range 0–10) designed to assess the completeness and rigour of reporting practices. The MRQS captures fundamental aspects of methodological reporting endorsed in educational research guidelines [24, 86, 96, 203] and codable from published study reports.

The ten MRQS items are:

1. **M1: Sample size explicitly reported** – transparent reporting of participant numbers as a minimum requirement for evaluating statistical conclusions [203];
2. **M2: Sample size ≥ 30** – a commonly regarded minimum for basic parametric inference [37];
3. **M3: Control or comparison group present** – necessary for causal inference about intervention effects [96, 203];
4. **M4: Validated measurement instruments used** – prerequisites for trustworthy outcome assessment;
5. **M5: Effect sizes or standardised measures reported** – enabling cross-study comparison, recommended by APA and CONSORT guidelines [18, 95, 203];
6. **M6: Pre-post or longitudinal design** – permitting assessment of within-participant change;
7. **M7: Inferential statistics reported** – providing probabilistic evidence regarding

the reliability of observed differences;

8. **M8: Explicit definition of gamification provided** – necessary for construct validity and cross-study comparability [20, 54];
9. **M9: Theoretical framework explicitly stated** – situating findings within explanatory frameworks;
10. **M10: Study limitations explicitly discussed** – enabling readers to calibrate confidence in findings.

The MRQS is a measure of *reporting* quality, not of overall study merit [125]. A qualitative study could score low because several items (M2, M3, M5, M6, M7) are not applicable to its design. This limitation is acknowledged in the interpretation of results.

In addition to the MRQS, risk-of-bias assessment was conducted using an adapted version of the ROBIS tool [241], modified for individual study appraisal. The adapted tool comprised three domains, each with two signalling questions:

1. **Domain 1 – Participant selection and study context:** (Q1) Are the criteria for selecting participants clearly described? (Q2) Is the sample representative of the target population?
2. **Domain 2 – Data collection and analysis:** (Q3) Were valid and reliable tools used for data collection? (Q4) Are the methods of data analysis described and justified?
3. **Domain 3 – Interpretation and reporting:** (Q5) Are interpretations and conclusions consistent with results? (Q6) Are limitations acknowledged and discussed?

Domain-level judgments followed three rules: (1) both signalling questions “yes” → low risk; (2) at least one “no” → high risk; (3) at least one “unclear” with no “no” → unclear risk.

1.3.3. Quality assessment results

1.3.3.1. Overall MRQS distribution

Across the 74 included studies, the mean MRQS was 4.31 out of 10 (SD = 1.93, Mdn = 4.0, IQR = 3–6, range = 0–9) [125]. No study achieved the maximum score of 10. The distribution was approximately unimodal: 15 studies (20.3%) scored in the low range (0–2), 38 (51.4%) in the middle range (3–5), and 21 (28.4%) in the upper range (6–9). The internal consistency was moderate (Cronbach's $\alpha = 0.58$), consistent with the instrument's design as a multi-dimensional reporting checklist rather than a single-construct scale.

1.3.3.2. Item-level prevalence

Table 1.11 presents the prevalence of each MRQS item, revealing three tiers of reporting quality.

Table 1.11: MRQS item prevalence across 74 studies, with temporal comparison

Item	Criterion	All (n=74)	2004–17	2018–24	Δ
M9	Theoretical framework	70 (94.6%)	91.9%	97.3%	+5.4
M1	Sample size reported	62 (83.8%)	78.4%	89.2%	+10.8
M2	Sample ≥ 30	37 (50.0%)	51.4%	48.6%	–2.7
M10	Limitations discussed	35 (47.3%)	37.8%	56.8%	+18.9
M6	Pre-post design	29 (39.2%)	43.2%	35.1%	–8.1
M7	Inferential statistics	29 (39.2%)	37.8%	40.5%	+2.7
M3	Control group	23 (31.1%)	37.8%	24.3%	–13.5
M4	Validated instruments	16 (21.6%)	18.9%	24.3%	+5.4
M8	Explicit definition	11 (14.9%)	18.9%	10.8%	–8.1
M5	Effect sizes reported	7 (9.5%)	8.1%	10.8%	+2.7

High-prevalence items (>75%). Nearly all studies included a theoretical framework (94.6%), and most reported sample sizes (83.8%), indicating that basic framing and descriptive practices are well-established.

Mid-prevalence items (20–50%). Half the studies had sample sizes ≥ 30 . Limitations discussion showed the greatest improvement over time (+18.9 percentage points).

Notably, the use of control groups *declined* from the earlier to the later period (–13.5 pp), reflecting the compositional shift toward qualitative and design-based studies.

Low-prevalence items (<20%). Only 21.6% used validated instruments, only 14.9% provided explicit definitions [43, 45, 88, 106, 134, 135, 165, 180, 201, 208, 252], and only 9.5% reported effect sizes. The last finding is particularly consequential: more than 90% of studies – including those reporting inferential statistics – did not report standardised effect measures, severely limiting meta-analytic synthesis [18, 95, 225].

1.3.3.3. MRQS by study characteristics

Journal articles scored slightly higher than conference papers (mean 4.75 vs. 4.17, ns), and studies providing explicit definitions scored higher than those without (mean 5.36 vs. 4.13, $p = .065$) [125]. Studies rated as having high overall risk of bias scored lower on the MRQS (mean 4.13) than those rated moderate or lower (mean 5.25).

1.3.3.4. Risk-of-bias results

The overall risk-of-bias assessment revealed [125, 241]:

- High risk of bias: 62 studies (83.8%)
- Moderate risk of bias: 9 studies (12.2%)
- Low risk of bias: 1 study (1.4%)
- Unclear: 2 studies (2.7%)

Domain-level analysis:

Domain 1 – Participant selection and study context was the most persistent source of weakness, with 85.1% of studies rated as high risk [125]. The signalling question on sample representativeness (Q2) was the single weakest indicator across the entire instrument: no study received a favourable rating on this criterion. This reflects the universal reliance on convenience sampling (intact classes, available groups) without

discussion of how the sample relates to a broader population [203].

Domain 2 – Data collection and analysis showed 63.5% of studies at high risk. The primary concerns were the use of unvalidated instruments (only 21.6% reported validity or reliability evidence) and insufficient description of analytical procedures.

Domain 3 – Interpretation and reporting was the strongest domain, with 55.4% at high risk. Studies generally aligned conclusions with results (Q5), but the discussion of limitations (Q6) was met by fewer than half the studies.

1.3.4. LLM validation of quality assessment

The parent systematic review employed LLMs at multiple stages: Claude 3.5 Sonnet for eligibility screening, and GPT-4o for data extraction and risk-of-bias assessment [125, 148, 168, 226]. To evaluate the reliability of these automated assessments, we conducted a systematic comparison.

Risk-of-bias assessment agreement. Comparison of GPT-4o’s automated risk-of-bias responses against human-verified judgments across the six signalling questions for all 74 studies revealed only 37.6% overall agreement [125]. Cohen’s κ values ranged from -0.04 to 0.21 across the signalling questions, corresponding to “poor” to “slight” agreement according to the Landis and Koch [131] benchmarks [37, 67]. The model consistently applied less stringent criteria than human reviewers, tending to rate studies more favourably on sampling quality (Q1, Q2) and instrument validation (Q3) [197].

These low agreement rates contrast markedly with the higher reliability observed for factual data extraction.

Factual metadata extraction. A supplementary multi-LLM comparison involving Gemini 3 Pro, ChatGPT-4, and Grok 4.1-thinking – each independently extracting data using the same 21-item prompt – achieved substantially higher agreement [125]. Pairwise agreement on objective, verifiable fields (sample size, year, publication type,

country) ranged from 82% to 97%, demonstrating that LLMs are highly reliable for factual extraction but unreliable for subjective quality judgments [148, 197, 226].

Implications for automated research synthesis. The divergent reliability patterns have important methodological implications [125]. LLMs can effectively accelerate the labour-intensive process of data extraction in systematic reviews (where they achieve near-human accuracy on factual items) [148]. However, they cannot reliably replace human judgment for risk-of-bias assessment, where the evaluative component requires nuanced interpretation of study design features, statistical appropriateness, and reporting completeness [197, 226]. The finding that GPT-4o applied less stringent criteria than human reviewers is particularly concerning, as it means that LLM-assisted reviews may systematically underestimate the risk of bias in the included studies.

These findings suggest a hybrid approach: LLMs for initial screening and factual extraction, with human verification for all evaluative judgments. This approach was, in fact, the method employed in the parent systematic review, and its success validates the hybrid model for educational research synthesis.

1.3.5. Definitional practices and quality

The relationship between definitional practices and methodological rigour reveals an important finding [125]. Studies providing explicit gamification definitions [43, 45, 88, 106, 134, 135, 165, 180, 201, 208, 252] scored higher on the MRQS (mean 5.36 vs. 4.13, $p = .065$) but showed equivalent risk-of-bias ratings. This suggests that definitional clarity functions as a marker of reporting conscientiousness – authors who take the time to define their key construct also tend to report other methodological details more thoroughly – rather than as a direct indicator of design rigour. A study can define gamification clearly while still employing a small sample and no control group.

The decline in definitional clarity from the earlier to the later period (–8.1 percent-

age points) is counterintuitive: one might expect that definitional conventions would strengthen as a field matures. The opposite pattern may reflect the normalisation of gamification terminology – as the concept becomes familiar, authors may assume that definitions are unnecessary, despite the continued conceptual ambiguity documented in this review.

The inter-item analysis revealed a quantitative-methods cluster (M3 control group, M6 pre-post design, M7 inferential statistics, with phi correlations of 0.36–0.60) operating independently of the conceptual clarity items (M8 definition, M9 framework) [125]. This structural finding suggests that the methodological quality of gamification research operates along two independent dimensions: *empirical design quality* (related to study design, measurement, and analysis) and *conceptual reporting quality* (related to definitions, frameworks, and limitations). Improving the field requires attention to both dimensions simultaneously [34, 56].

1.3.6. Detailed bias analysis by category

Beyond the aggregate statistics, the risk-of-bias assessment revealed specific patterns of methodological weakness that merit detailed examination.

1.3.6.1. *Sample size limitations*

A substantial proportion of included studies exhibited significant risk of bias due to inadequate sample sizes [125]. Of the 62 studies reporting participant numbers, 41 (66.1%) utilised samples below 50 participants, raising concerns about statistical power and reliability [37]. Studies with particularly small samples ($n \leq 17$) included VR-based studies ($n = 17$) [9], first-year student evaluations ($n = 5$) [167], respondent surveys ($n = 5$), preliminary evaluations ($n = 4$) [7, 71], and teacher observations with classroom observations ($n = 8$ teachers). Several studies explicitly acknowledged sample size as a

limitation, noting “limited sample size” as a constraint or identifying insufficient sample size for robust quantitative analysis.

1.3.6.2. Research design concerns

Significant heterogeneity in research designs contributed to varying levels of bias risk [96, 125]. Approximately 25 studies (31%) lacked comparison groups entirely, relying solely on pre-post single-group designs without controls. This design limitation precludes causal inference regarding the effects of interventions [203]. Several studies employed purely qualitative methodologies without quantitative outcome measures. While qualitative approaches provide valuable insights into user experience and implementation processes, the absence of standardised outcome metrics introduces subjectivity and limits cross-study comparability [34, 41].

Studies utilising quasi-experimental designs with non-random assignment employed convenience sampling or intact classroom groups, introducing potential selection bias and confounding variables. One study explicitly acknowledged possible instructor bias, as the same teacher delivered both experimental and control conditions, potentially influencing differential treatment effects.

1.3.6.3. Measurement and assessment issues

Numerous studies demonstrated risk of bias related to outcome measurement [125]. Some relied exclusively on perceived learning rather than objective assessments, utilising Likert scale questionnaires without validation of actual knowledge gains [76, 159]. Self-report bias was prevalent, with many studies relying primarily on self-reported data without objective verification [147, 247]. Instrument validation was frequently absent or inadequately reported: studies used unvalidated survey questionnaires with no reported reliability information, and game testing employed no standardised assessment

tools [62, 116].

1.3.6.4. Confounding variables and external validity

Several studies failed to adequately control for potential confounding variables, including prior gaming experience (which could substantially influence engagement and learning outcomes) [107, 245] and epistemological beliefs and prior game preferences. The novelty effect represented a recurring threat to validity, where student engagement might be artificially inflated due to the unfamiliarity of game-based approaches rather than their intrinsic educational value [56, 89].

Generalisability was severely limited in numerous studies. Single-institution studies could not account for variations in educational systems, cultural contexts, or institutional resources that might moderate intervention effects. Geographic and cultural constraints further limited external validity: the concentration of studies in a small number of countries (Taiwan, USA, Malaysia, Indonesia accounting for 55.4%) means that findings may not generalise to other educational contexts.

1.3.6.5. Implementation fidelity

Implementation fidelity issues introduced additional bias risks [125]. Studies reported significant technical difficulties including GPS/connectivity problems, hardware limitations, and AR recognition issues that interfered with intervention delivery [7, 194, 208]. Teacher-related factors represented another concern, with variability in implementation approaches: some teachers used games merely as visual aids rather than engaging students in active historical inquiry, potentially diluting intervention effects [115]. Time constraints affected multiple studies, with insufficient class time for adequate implementation leading to incomplete student experiences [107, 218].

1.3.7. Implications for the field

The high prevalence of methodological weaknesses has several important implications:

First, the positive findings regarding gamification effectiveness reported in Section 1.2 should be interpreted cautiously. Studies with high risk of bias may overestimate effect sizes. Previous meta-analyses in educational technology have shown that studies with higher methodological quality tend to report smaller effect sizes [92], suggesting that the true effectiveness of gamification in history education may be more modest than the current literature implies.

Second, the field urgently needs higher-quality research designs [125, 203]. The decline in control group usage (-13.5 pp from 2004–2017 to 2018–2024) is particularly concerning, as it means the newer literature is less capable of supporting causal inference than the older literature. Quasi-experimental designs with adequate sample sizes (≥ 64 per group for medium effects) and pre-registered protocols should become the standard for gamification effectiveness research [86].

Third, the definitional clarity issue is not merely a semantic concern but a methodological one [20, 54]. Without clear definitions, construct validity cannot be established, and it becomes impossible to determine whether different studies are investigating the same phenomenon [56]. The field would benefit from adopting a shared taxonomy that distinguishes structural gamification, content gamification, full gamification, and game-based learning [201, 205].

Fourth, the near-universal failure to report effect sizes (90.5% of studies) prevents meta-analytic synthesis and makes it impossible to estimate the practical significance of gamification effects [18, 95]. Future studies should routinely report Cohen's d , eta-squared, or equivalent measures alongside inferential test results [37].

Fifth, the LLM validation findings have important implications for the growing practice of AI-assisted systematic reviews [148, 226]. While LLMs can accelerate factual data extraction (82–97% agreement), they cannot replace human judgment for evaluative tasks like risk-of-bias assessment ($\kappa = -0.04$ to 0.21) [131]. Reviews that rely on LLM-only quality assessment may systematically underestimate methodological problems [197].

1.3.8. Proposed reporting checklist

Based on the MRQS analysis [125], we propose that future studies on gamification in history education should satisfy the following minimum reporting requirements:

1. Provide an explicit definition of gamification (or game-based learning, serious games, etc.) and position the study within the four-category taxonomy (structural, content, full, game-based learning) [20, 54, 201];
2. Report the sample size, sampling method, and any available demographic characteristics [203];
3. Describe the measurement instruments used and provide evidence of validity and/or reliability;
4. Report effect sizes alongside inferential statistics [18, 37];
5. Include a comparison condition (even if non-equivalent) where the research question involves effectiveness claims [96, 203];
6. Discuss study limitations explicitly, including threats to internal and external validity;
7. Specify the theoretical framework guiding the design and interpretation;
8. Report the duration of the intervention and the timing of assessments;
9. Describe the game design elements employed and the rationale for their selection [198, 205];

10. Make data and materials available where possible to support replication.

This checklist operationalises the MRQS items into actionable guidance for researchers. If adopted, it would address the most critical reporting gaps identified in the current literature and move the field toward the transparent, rigorous evidence base that practitioners need.

1.3.9. Conclusions to Section 1.3

The MRQS quality assessment [125] reveals that the vast majority (83.8%) of gamification studies in history education carry a high risk of bias. The mean MRQS was 4.31/10, with the most critical weaknesses being the near-universal failure to report effect sizes (90.5%) [18], the rarity of validated instruments (21.6%), and the pervasive definitional ambiguity (85.1%) [20, 54]. Domain-level risk-of-bias analysis [241] identified participant selection (Domain 1, 85.1% high risk) as the most persistent source of weakness, with no study meeting the representativeness criterion. The LLM validation study revealed that automated risk-of-bias assessment achieves only slight agreement with human reviewers ($\kappa = -0.04$ to 0.21) [67, 131], while factual data extraction is highly reliable (82–97% agreement) [148, 197, 226]. These findings collectively inform both the design of the experimental study (Chapter 2) and the self-assessment approach adopted in the discussion (Chapter 3).

1.4. Gamification and digital competence: implementation through DigComp 2.0

1.4.1. The DigComp 2.0 framework

The European Digital Competence Framework for Citizens (DigComp 2.0) provides a comprehensive reference framework for digital competence [238]. The framework identifies five competence areas:

1. **Information and data literacy:** Browsing, searching, and filtering data; evaluating data; managing data.
2. **Communication and collaboration:** Interacting through digital technologies; sharing through digital technologies; engaging in citizenship through digital technologies; collaborating through digital technologies; netiquette; managing digital identity.
3. **Digital content creation:** Developing digital content; integrating and re-elaborating digital content; copyright and licences; programming.
4. **Safety:** Protecting devices; protecting personal data and privacy; protecting health and well-being; protecting the environment.
5. **Problem solving:** Solving technical problems; identifying needs and technological responses; creatively using digital technologies; identifying digital competence gaps.

1.4.2. Mapping gamification to DigComp 2.0

The implementation of gamification in history education can be systematically aligned with DigComp 2.0 competence areas [232] [123]. Table 1.12 presents a detailed mapping of gamification activities to specific DigComp sub-competencies, demonstrating that gamified history education simultaneously develops historical knowledge and digital competencies [236].

Information and data literacy (Area 1). History quiz-based gamification directly engages students with information processing tasks: interpreting primary sources presented in digital format, evaluating the reliability of historical evidence, and managing data across multiple assessment forms [123, 238]. In the context of this study's 156 Google Forms quizzes, students practised data literacy with each assessment interaction. The multi-block structure of seminars (typically 3–7 quiz blocks per session) required

Table 1.12: Mapping of gamification activities to DigComp 2.0 competence areas

DigComp area	Sub-competencies	Gamification activities
1. Information and data literacy	1.1 Browsing, searching; 1.2 Evaluating; 1.3 Managing	Navigating Google Forms quizzes; evaluating historical sources in quiz questions; managing responses across 156 forms
2. Communication and collaboration	2.1 Interacting; 2.2 Sharing; 2.4 Collaborating; 2.5 Netiquette	Viber group interactions; sharing resources and quiz results; peer support; appropriate communication norms
3. Digital content creation	3.1 Developing; 3.2 Integrating	Interacting with History-LikeGame platform; engaging with multimedia course materials; creating responses in varied digital formats
4. Safety	4.1 Protecting devices; 4.3 Health and well-being	Wartime digital safety awareness; balancing online learning with physical safety; managing screen time
5. Problem solving	5.1 Technical problems; 5.2 Identifying needs; 5.3 Creative use	Solving quiz problems; adapting to technical disruptions; finding alternative access methods during outages

students to navigate between forms, track their progress, and manage their responses – activities that develop competencies in data management (sub-competency 1.3).

Communication and collaboration (Area 2). The Viber chat group used in the course served dual purposes: facilitating communication about course logistics and creating a space for collaborative learning [123, 238]. Students shared resources, discussed historical questions, and provided peer support. The gamified naming conventions for quiz blocks (using themes from Greek mythology, gemstones, flowers, and minerals) created a shared vocabulary that fostered community and demonstrated the social function of creative digital communication [84, 146]. The chat log analysis documented 400 messages reflecting various communication patterns: formal absence notifications, informal peer interaction, resource sharing, and instructor support – all contributing to sub-competency 2.5 (netiquette) and 2.4 (collaborating through digital technologies).

Digital content creation (Area 3). The gamified platform HistoryLikeGame engaged students with interactive digital content including drag-and-drop matching exercises and text-input challenges [123]. While students were primarily consumers rather than creators of digital content in this study, the platform’s HTML-based architecture demonstrated how educational digital content can be created and deployed using standard web technologies [238]. The open-source nature of the platform (hosted on GitHub Pages) provides a model for educators interested in developing their own gamification tools, addressing sub-competency 3.1 (developing digital content).

Safety and well-being (Area 4). The wartime context of this study added a critical and unique dimension to digital safety that is absent from most DigComp implementations [123, 238]. Students navigated online learning while managing real threats including power outages, internet disruptions, and air raid alerts. The course’s asynchronous components and archived materials ensured continued access to education even when synchronous participation was impossible. The instructor’s explicit safety guidance (“Your safety is the most important [...] if in doubt, go to a shelter”) modelled responsible digital behaviour that prioritises physical well-being over digital participation – a competency area (4.3: protecting health and well-being) that takes on heightened significance in conflict zones [78].

Problem solving (Area 5). Gamified assessments inherently involve problem-solving: students must identify relevant information, apply historical knowledge to novel contexts, and select appropriate strategies for different question types (matching, sequencing, multiple choice, open-ended) [123, 174, 238]. The wartime context further developed problem-solving competencies (sub-competency 5.1: solving technical problems) as students adapted to infrastructure disruptions by finding alternative ways to access course materials and complete assessments.

1.4.3. Implementation guidelines

Based on the synthesis of the systematic reviews and the DigComp 2.0 alignment analysis [123, 238], the following guidelines for implementing gamification in history education were developed. Each guideline is grounded in the evidence base established in Sections 1.1–1.3 and is connected to specific DigComp competence areas.

1. **Start with clear learning objectives** (DigComp Areas 1, 5): Define specific historical knowledge outcomes and digital competencies to be developed before selecting gamification elements. The MRQS analysis showed that studies with explicit theoretical frameworks (94.6%) generally produced more interpretable results. The learning objectives should specify: (a) the historical content to be mastered, (b) the historical thinking skills to be developed, and (c) the digital competencies to be practised.
2. **Use progressive complexity** (DigComp Area 5): Structure gamified activities in a sequence of increasing difficulty, from recognition tasks (matching, multiple choice) through analysis tasks (source evaluation, chronological sequencing) to synthesis tasks (constructing narratives, arguing causation). This progression mirrors the cognitive development sequence identified in the design principles (Section 1.2) and aligns with established taxonomies of historical thinking.
3. **Maintain historical rigour** (DigComp Area 1): Ensure that game elements do not trivialise or distort historical content. The systematic review found that 10.8% of studies reported negative outcomes, often associated with superficial gamification that prioritised engagement over accuracy. All quiz questions should be based on verified historical facts and aligned with curriculum standards (in this study, the NMT Ukrainian History syllabus).
4. **Provide multiple pathways** (DigComp Areas 3, 5): Offer both required and op-

tional (“extra”) gamified activities to accommodate different ability levels and engagement preferences. The HistoryLikeGame platform exemplifies this approach with its five sections (plus Omega comprehensive review), allowing students to choose their level of engagement. Wordwall and other supplementary platforms provide additional variety.

5. **Integrate assessment with learning** (DigComp Areas 1, 5): Use gamified quizzes as both formative assessment tools and learning activities, providing immediate feedback and opportunities for retry. The case of Talash D. (Chapter 3), who submitted 7 attempts at a comprehensive test with progressively improving scores, demonstrates the power of this approach for iterative mastery learning.
6. **Ensure accessibility and resilience** (DigComp Area 4): Design for various devices and connectivity conditions, particularly important in contexts where infrastructure may be unreliable. The HistoryLikeGame platform’s static HTML architecture (total size: 205 KB) provides a model for accessible design. In wartime or disaster-affected contexts, the five-layer resilience model (synchronous, communication, content, assessment, gamification) ensures educational continuity.
7. **Build community through gamification** (DigComp Area 2): Use creative naming, shared references, and collaborative elements to foster a sense of belonging and shared purpose among learners. The thematic quiz naming conventions (gemstones, Greek mythology, flowers, trees, minerals) served this function in the present study, creating cultural enrichment opportunities alongside the history content.
8. **Document systematically** (DigComp Areas 1, 3): Maintain detailed records of implementation (chat logs, assessment data, platform analytics) to enable evidence-based evaluation and future research. The MRQS analysis revealed that compre-

hensive documentation is the exception rather than the rule in gamification research – only 47.3% of studies discussed limitations, and only 9.5% reported effect sizes. Systematic documentation addresses the most critical reporting gaps identified in Section 1.3.

These guidelines synthesise theoretical insights from the four analyses conducted in this chapter (digital history review, gamification review, MRQS assessment, and DigComp alignment) into actionable recommendations for practitioners. Their application in the experimental study is documented in Chapter 2.

1.4.4. Conclusions to Section 1.4

The alignment of gamification with DigComp 2.0 [238] demonstrates that gamified history education can simultaneously develop historical knowledge and all five areas of digital competence: information and data literacy, communication and collaboration, digital content creation, safety, and problem solving [123, 232]. The wartime context of this study adds a unique dimension to the safety competence area, as students must navigate real physical threats alongside digital learning challenges [78].

The eight implementation guidelines derived from the synthesis of systematic reviews [124–126] provide a practical framework for educators seeking to implement gamification in their history courses. These guidelines address the key weaknesses identified in the MRQS assessment (lack of clear objectives, insufficient documentation, absent definitions) [125] while operationalising the six design principles from the gamification review (historical authenticity, narrative integration, progressive difficulty, immediate feedback, social elements, multiple modalities) [126].

1.5. Conclusions to Chapter 1

The theoretical analysis presented in this chapter establishes the following key findings:

1. **Digital history pedagogy is an emerging field:** The PRISMA-guided systematic review [124, 177] identified only 6 pedagogical studies [12, 39, 52, 133, 153, 213], indicating that digital history in education lacks the critical mass of research needed for evidence-based practice (Section 1.1).
2. **Gamification is the most researched digital approach to history education:** The systematic review of 74 studies [126] confirms that gamification has attracted the most research attention among digital history methods, but the field is characterised by definitional ambiguity (85.1% lack definitions) [54] and predominantly positive but methodologically weak evidence (Section 1.2).
3. **Methodological quality is critically low:** The MRQS assessment [125] reveals that 83.8% of gamification studies in history education carry a high risk of bias, primarily due to small samples, lack of randomisation, and insufficient reporting. The LLM validation approach [148, 197, 226] offers a promising pathway for scaling quality assessment (Section 1.3).
4. **Gamification can develop digital competencies:** Systematic alignment with DigComp 2.0 [238] demonstrates that gamified history education activities address all five competence areas [123, 232], making gamification not merely an engagement tool but a vehicle for comprehensive digital competence development (Section 1.4).
5. **Evidence-based guidelines are needed:** The synthesis of findings from all four analyses [123–126] provides the foundation for implementation guidelines that address the identified gaps: clear definitions [20, 54], rigorous design [203], pro-

gressive complexity, and systematic documentation.

These findings collectively justify the experimental study presented in the following chapters: a gamification-enhanced digital history course designed according to the identified principles [125, 126], implemented in a real educational setting, and documented with sufficient detail to address the methodological weaknesses prevalent in the field [125, 203].

CHAPTER 2

EXPERIMENTAL METHODOLOGY

2.1. Organisation of the pedagogical experiment

2.1.1. Institutional context

The pedagogical experiment was conducted at the Centre for Pre-University Training and Education (CDPO) of Kryvyi Rih State Pedagogical University (KDPU), Ukraine [128]. CDPO offers preparatory courses for prospective university applicants preparing for the National Multi-Subject Test (NMT) – Ukraine’s standardised university entrance examination [141]. The Ukrainian History component of the NMT covers the entire school curriculum from ancient times to the present day, requiring comprehensive knowledge across all historical periods.

The course was delivered entirely via distance learning [97, 249], reflecting both the institutional adoption of online education since the COVID-19 pandemic and the ongoing security situation due to Russia’s full-scale invasion of Ukraine, which began on February 24, 2022 [150, 151, 237]. The city of Kryvyi Rih, located in Dnipropetrovsk Oblast, experienced regular missile and drone attacks throughout the study period, directly affecting participants’ ability to attend synchronous sessions [249].

2.1.2. The National Multi-Subject Test (NMT)

The NMT replaced the earlier ZNO (External Independent Evaluation) system beginning in 2022, adapting Ukraine’s university entrance assessment to wartime conditions [60]. The NMT for Ukrainian History encompasses the complete school history curriculum and requires competencies in:

- **Chronological knowledge:** Sequencing events across all historical periods from

ancient times to the present;

- **Factual recall:** Identifying key figures, dates, treaties, and events;
- **Source analysis:** Interpreting primary sources including texts, maps, images, and statistical data;
- **Cause and effect:** Explaining causal relationships between historical events and processes;
- **Comparative analysis:** Identifying similarities and differences across historical periods and regions.

The test format includes multiple-choice questions, matching/correspondence tasks, chronological sequencing, and open-ended items [141]. Table 2.1 summarises the NMT question types and their correspondence to the gamification elements used in this study.

Table 2.1: NMT question types and corresponding gamification elements

NMT question type	Cognitive demand	Gamification correspondence
Multiple choice	Recognition, factual recall	Google Forms quiz blocks
Matching/correspondence	Association, categorisation	HistoryLikeGame drag-and-drop
Chronological sequencing	Temporal reasoning	HistoryLikeGame drag-and-drop
Source analysis	Interpretation, evaluation	Seminar source-based questions
Open-ended	Recall, synthesis	HistoryLikeGame text input

The 2025 NMT was scheduled for May 14, 2025, providing the external deadline that shaped the course timeline and influenced student engagement patterns [238]. The proximity of this deadline explains the observed increase in participation in later seminars (Section 3).

Figure 2.1 illustrates the visual source analysis component of the course materials, showing the type of historical portrait identification exercises that correspond to the NMT source analysis question format.



Figure 2.1: Course material: visual source identification exercise featuring historical portraits of key Ukrainian figures (Mazepa, Hrushevskyi, Petliura, Stalin) with associated artifacts

2.1.3. Participants

Twenty-three students enrolled in the NMT preparatory course for Ukrainian History during the study period. Enrolment was open and occurred on a rolling basis, with the complete enrollment timeline reconstructed from the Viber chat log join events (Table 2.2).

Table 2.2: Detailed enrollment timeline from chat log analysis

No.	Date	Phase	Cumulative
1	2024-10-01	Phase 1	1
2	2024-10-04	Phase 1	2
3–6	2024-10-06–08	Phase 1	6
7–9	2024-10-09–11	Phase 1	9
10	2024-10-16	Phase 1	10
11	2024-10-25	Phase 1	11
12–17	2024-11-01–13	Phase 2	17
18–19	2025-01-22–27	Phase 3	19
20–21	2025-02-13–25	Phase 3	21
22	2025-03-03	Phase 3	22
23	2025-04-11	Phase 3	23

The three enrollment phases reflect distinct student motivations:

Phase 1 (October 1–25, 2024): Eleven students joined within the first three weeks, forming the core group. This phase coincided with the announcement of course schedules through CDPO administrative channels and represents students who made early decisions to prepare for the NMT.

Phase 2 (November 1–13, 2024): Six additional students enrolled during November, bringing the total to 17. These students may have been motivated by word-of-mouth recommendations from Phase 1 participants or delayed administrative processing of their applications.

Phase 3 (January 22 – April 11, 2025): Six more students joined during the second half of the course. This late enrollment included the final participant who joined on April 11, 2025 – just 33 days before the NMT examination. Late enrollment is common

in CDPO preparatory courses and may reflect students who initially planned self-study but later decided that structured preparation was necessary.

The rolling enrollment pattern is typical for CDPO preparatory courses and reflects the open-access nature of the programme. Participants were predominantly secondary school graduates or final-year students preparing for the 2025 NMT examination [60]. Demographic data were limited by the course format; individual identification was possible through the names provided in Google Forms responses and the Viber chat group.

Ethical considerations: As the study employed an ex post facto design [8, 46] analysing data collected during regular educational activities (not designed as research), no separate informed consent for research participation was required. All data were anonymised in reporting, with student names used only for internal trajectory analysis and replaced with initials or pseudonyms in publications.

2.1.4. Research design

The study employed an ex post facto single-group repeated-measures design [8, 46]. This design was appropriate given:

1. The educational context did not permit random assignment to treatment and control groups, as all enrolled students were entitled to the same educational services [199];
2. The primary interest was in documenting the implementation process and tracking performance over time within a real educational setting [192];
3. The wartime context introduced variables that could not be controlled experimentally but constituted important contextual factors [78, 150].

This design aligns with the acknowledged need in the field (Section 1.3) for studies that prioritise ecological validity and comprehensive documentation over artificial

experimental control [56, 89, 125].

2.1.5. Timeline

The course spanned approximately six months, from October 12, 2024, to April 13, 2025 (the last recorded session in available data). The following phases were identified:

Phase 1: Baseline (October 12 – November 3, 2024). The course commenced with introductory lectures and an introductory test to establish baseline knowledge. Three seminars (1–3) and the first modular control were administered. The Zoom video conferencing platform served as the primary synchronous teaching tool [97].

Phase 2: Formative stage without gamification (November 10 – November 30, 2024). Regular instruction continued with lectures 6–10 and seminars 4–8 [16]. The course established its rhythm of Sunday sessions combining lectures with seminar quizzes. The second modular control was administered on December 1, 2024.

Phase 3: Introduction of gamification (December 1, 2024). The gamified platform HistoryLikeGame was introduced as a supplementary resource, implementing gamification as the use of game design elements in non-game contexts [54]. Students were encouraged to explore the “Historical Quest” section. Additional digital tools for history education, including VR/AR applications [5] and interactive multimedia [98], were considered but not deployed due to bandwidth constraints in the wartime context. The Wordwall platform was later introduced (March 27, 2025) as an additional gamification tool.

Phase 4: Formative stage with gamification (December 2024 – April 2025). The course continued with an enriched methodology integrating traditional lectures, Google Forms seminars, modular controls, and the gamified platform [124, 232]. Additional weekday sessions were organised in December to compensate for schedule disruptions [150]. Five further modular controls (3–7) were administered.

2.1.6. Course structure

The course comprised five integrated instructional components [16, 68, 225], summarised in Table 2.3:

Table 2.3: Overview of course components

Component	Count	Mode	Assessment
Lectures	38	Synchronous (Zoom)	Video recordings shared
Seminars	30	Quiz-based (Google Forms)	133 quiz blocks
Modular controls	7	Mandatory (Google Forms)	7 control forms
Gamification modules	23	Self-paced (HTML)	Self-assessment
Supplementary	–	Asynchronous (Drive)	Worksheets, PDFs

Figure 2.2 shows the gamified NMT structure poster used to introduce students to the course format.

Lectures. The 38 lectures covered Ukrainian history from ancient times to the present, organised into four thematic sections corresponding to the HistoryLikeGame platform structure:

1. **Section 1** (Lectures 1–10): Ancient and medieval Ukrainian history – from the earliest settlements through Kyivan Rus', the Galicia-Volhynia Kingdom, and the Lithuanian-Polish period;
2. **Section 2** (Lectures 11–18): Early modern period – the Cossack era, Hetmanate, Khmelnytsky rebellion, Zaporizhian Sich, and the incorporation into the Russian and Habsburg empires;
3. **Section 3** (Lectures 19–28): 19th and early 20th century – national awakening, cultural movements, World War I, the Ukrainian revolution (1917–1921), and the interwar period;
4. **Section 4** (Lectures 29–38): 20th and 21st century – World War II, Soviet Ukraine, the dissident movement, independence (1991), and contemporary Ukrainian history.



Figure 2.2: Gamified NMT structure poster (“Know the Test”) used in the course to introduce the NMT examination format through engaging visual design

Each lecture was delivered via Zoom with screen sharing of presentation slides. All lectures were recorded and the videos were shared through the Viber chat group, ensuring asynchronous access for students unable to attend live sessions [16, 225]. Supplementary worksheets were distributed via Google Drive and direct chat file sharing.

Seminars. The 30 seminars were the core formative assessment component [92, 94], each comprising multiple Google Forms quiz blocks (typically 3–7 blocks per seminar, as detailed in Appendix A). The seminar structure evolved throughout the course:

- Early seminars (1–8) used simple numbering (“Блок 1”, “Блок 2”) or Roman numerals;
- From Seminar 17 onwards, creative thematic naming was consistently applied (see Section 2.2.2), representing a structural gamification element [198, 205];
- Quiz complexity increased progressively, from basic recall questions in early seminars to comprehensive NMT-format assessments in later seminars (e.g., Seminar 37 based on the 2016 NMT examination) [34].

Modular controls. Seven mandatory modular controls were administered at regular intervals, each covering the content of 5–8 preceding lectures [92]. The mandatory nature of these assessments (emphasised in the chat log with phrases like “ОБОВ’ЯЗКОВО до виконання” – “Mandatory for completion”) distinguished them from the voluntary seminar quizzes and provided standardised checkpoints across the course.

Total assessment volume. The 156 Google Forms links identified in the chat log represent a substantial assessment workload [92]. Assuming each form contained an average of 5–10 questions, the total question pool across all forms may be estimated at 780–1,560 individual questions, providing comprehensive coverage of the Ukrainian history curriculum.

2.2. Assessment tools and data collection

2.2.1. Google Forms quizzes

Google Forms served as the primary assessment and data collection platform [78, 97]. Each quiz was designed with the following characteristics:

- Questions aligned with the NMT examination format, including multiple choice, matching (correspondence), chronological sequencing, and source analysis;
- Automatic scoring with the “X / Y” format (points achieved / maximum points) [92];
- Immediate feedback upon submission, supporting the testing effect whereby retrieval practice strengthens memory [41];
- Timestamp recording for each response;
- Optional name field (introduced from Seminar 26 onwards, with “Напишіть своє прізвище та ім’я” – “Write your surname and name”).

The 156 quiz forms were distributed as follows (extracted from the chat log analysis):

Table 2.4: Distribution of Google Forms quizzes by type

Type	Count	Percentage
Seminar quizzes	133	85.3%
Lecture quizzes	7	4.5%
Modular controls	6	3.8%
Other (incl. intro test)	10	6.4%
Total	156	100%

2.2.2. Creative quiz naming conventions

A distinctive feature of the course’s gamification approach [54, 122] was the use of creative, thematic naming for quiz blocks within each seminar. Rather than using generic labels (“Block 1”, “Block 2”), the instructor adopted thematic naming schemes that changed with each seminar, creating an element of surprise and cultural enrich-

ment [198]. The naming themes included:

- **Gemstones:** Seminar 19 (Аметист/Amethyst, Берил/Beryl, Варисцит/Variscite, Геліодор/Heliodor)
- **Greek mythology:** Seminar 20 (Ахіл/Achilles, Беллерофонт/Bellerophon, Скамандр/Диомед/Diomedes, Гектор/Hector)
- **Greek alphabet:** Seminars 17, 21 (alpha, beta, gamma, delta, epsilon)
- **Flowers:** Seminars 25, 29 (Айстра/Aster, Барвінок/Periwinkle, Волошка/Cornflower, Гіацинт/Hyacinth)
- **Trees:** Seminar 27 (Акація/Acacia, Баобаб/Baobab, Вільха/Alder, Гінкго/Ginkgo)
- **Minerals:** Seminar 30 (Азбест/Asbestos, Берил/Beryl, Воксит/Bauxite, Галіт/Halite)
- **Roman numerals:** Seminar 8 (I, II, III, IV, V)
- **Chapter numbering:** Seminar 11 (Chapter 1–5)
- **Simple numbering:** Various seminars (Блок 1–5)

This naming practice exemplifies content gamification [205] – using creative framing to enhance the learning experience without modifying the assessment content itself [124, 231]. Students frequently expressed positive reactions to the naming schemes in the chat group.

2.2.3. Viber chat log

The Viber messenger group served as the primary communication channel for the course [237]. The chat log was exported and analysed, yielding 400 parsed messages over the study period. The chat data provided:

- **Enrollment timeline:** 28 join events for 23 unique participants (some joined multiple times after being removed and re-added);
- **Absence reports:** 33 messages reporting inability to attend, with classified reasons;

- **Assessment links:** All 156 Google Forms links shared during the course;
- **Teaching materials:** 51 file shares (worksheets, presentations) and 46 video shares (lecture recordings);
- **Gamification introductions:** Links to HistoryLikeGame platform (December 1, 2024; December 29, 2024) and Wordwall (March 27, 2025);
- **Wartime context:** Messages about power outages, internet disruptions, air raid alerts, and safety advisories [78, 150].

2.3. Gamification design: the HistoryLikeGame platform

2.3.1. Platform overview

HistoryLikeGame is an interactive web-based gamification platform developed specifically for this study [124, 231]. The platform is hosted at astrafreeman.github.io/HistoryLikeGame and consists of static HTML pages that can be accessed from any device with a web browser, requiring no software installation or account creation [235, 248]. Figure 2.3 shows the platform interface for a representative module.

2.3.2. Structure and content

The platform comprises 23 game modules organised into 5 sections (Table 2.5), reflecting the principle of progressive difficulty [34, 41]. The section structure mirrors the chronological organisation of the NMT curriculum:

Table 2.5: Structure of the HistoryLikeGame platform

Section	Modules	Content focus
Section 1	7	Ancient and medieval Ukrainian history
Section 2	5	Early modern period
Section 3	5	19th–early 20th century
Section 4	5	20th century
Omega	1	Comprehensive review
Total	23	



Figure 2.3: HistoryLikeGame platform interface: Section 2, Module 3 (S. Palia Uprising) showing drag-and-drop historical content and the Order of St. Vladimir visual element

The unequal distribution of modules across sections (7 for Section 1 vs. 5 for Sections 2–4) reflects the greater complexity and breadth of ancient and medieval Ukrainian history, which encompasses multiple civilisations, state formations, and cultural epochs [123]. Section 1 includes two supplementary modules (S_1_3_1 and S_1_5_1) that provide additional practice on topics identified as particularly challenging for students.

2.3.3. Question types and interaction design

Automated analysis of the 23 HTML source files (using the `catalog_gamification.py` module of the data pipeline) revealed two primary interaction types with distinct cognitive demands [34, 120, 184]:

- **Drag-and-drop** (present in 21 of 23 modules, 91.3%): Students match historical events, dates, or figures by dragging elements to correct positions. This format is particularly effective for chronological sequencing and correspondence tasks that

appear on the NMT [130, 188]. The drag-and-drop interaction engages recognition memory and spatial reasoning, requiring students to evaluate multiple options simultaneously [16].

- **Text input** (present in all 23 modules, 100%): Students type answers directly, requiring recall rather than recognition [16]. This more demanding format was used for key dates, names, and terminology. Text input questions assess active recall, which research in cognitive psychology has shown to produce stronger long-term retention than recognition-based tasks [34, 41].

Table 2.6 presents the technical characteristics of each module as extracted from the HTML analysis.

Table 2.6: Technical characteristics of HistoryLikeGame modules

Module	Section	Types	Buttons	Size (KB)
S_1_1	1	D&D + Text	9	8.6
S_1_2	1	D&D + Text	4	9.8
S_1_3	1	D&D + Text	4	9.7
S_1_3_1	1	D&D + Text	4	9.9
S_1_4	1	D&D + Text	4	9.0
S_1_5	1	Text only	7	6.4
S_1_5_1	1	D&D + Text	6	10.4
S_2_1	2	Text only	9	7.3
S_2_2	2	D&D + Text	4	10.0
S_2_3	2	D&D + Text	4	10.0
S_2_4	2	D&D + Text	9	8.1
S_2_5	2	D&D + Text	4	9.5
S_3_1	3	D&D + Text	9	8.1
S_3_2	3	D&D + Text	4	9.9
S_3_3	3	D&D + Text	4	9.5
S_3_4	3	D&D + Text	9	7.9
S_3_5	3	D&D + Text	9	8.3
S_4_1	4	D&D + Text	9	8.3
S_4_2	4	D&D + Text	4	11.1
S_4_3	4	D&D + Text	4	10.1
S_4_4	4	D&D + Text	9	8.1
S_4_5	4	D&D + Text	9	7.8
Omega	–	Text only	9	7.3
Mean size				8.9
Total size				205.1

The total platform size of approximately 205 KB makes it exceptionally lightweight

compared to other educational technology platforms [248]. For comparison, a single lecture recording typically occupies 100–500 MB, making the gamification platform approximately 500–2,500 times smaller. This size advantage is critical in the wartime context, where bandwidth limitations frequently affect educational access [150, 237].

Each module provides immediate visual feedback upon interaction, using colour coding (green for correct, red for incorrect) and score tracking [92]. The feedback mechanism aligns with the “immediate feedback” design principle identified in the systematic review (Section 1.2) [124].

2.3.4. Design rationale

The platform’s design was guided by the principles identified in Section 1.2 [124, 127]:

- **Historical authenticity:** All content was based on the NMT curriculum and verified historical facts [31, 124, 126];
- **Progressive complexity:** Sections follow the chronological structure of the course, with the Omega section providing a comprehensive challenge [34, 89, 132];
- **Accessibility:** Static HTML pages ensure minimal bandwidth requirements and universal device compatibility – critical in the wartime context where internet connectivity is unreliable [238, 248];
- **Immediate feedback:** Instant visual feedback supports self-regulated learning [41, 92];
- **Low barrier to entry:** No registration, login, or software installation required [184].

2.3.5. Alignment with design principles

The HistoryLikeGame platform was designed to implement the six evidence-based design principles identified in the systematic review (Section 1.2) [124, 126]. Table 2.7

maps each principle to specific platform features.

Table 2.7: Alignment of HistoryLikeGame platform with evidence-based design principles

Design principle	Platform feature	Evidence from review
Historical authenticity [124]	All content verified against NMT curriculum; factual accuracy maintained	Studies with accurate content reported stronger outcomes; 10.8% reporting negatives linked to trivialisation [186]
Narrative integration [198]	Chronological structure following historical periods; thematic quiz naming	Quest-based structures aligned with historical investigation; strongest effects on affective outcomes [126]
Progressive difficulty [34]	Four sections + Omega review; increasing complexity within sections	Scaffolded challenges from recognition to analysis to synthesis supported sustained engagement [185]
Immediate feedback [92]	Colour-coded responses (green/red); instant score display	Enabled iterative learning cycles; particularly effective for factual recall [250]
Social elements [188]	Viber group sharing; peer support; community naming	Collaborative structures showed more consistently positive effects than competition [89]
Multiple modalities [225]	Drag-and-drop + text input; visual + textual elements	Multimodal engagement addressed diverse learning preferences [41]

2.3.6. Supplementary gamification tools

In addition to HistoryLikeGame, the course utilised multiple gamification modalities [54, 205]:

- **Wordwall:** An external platform for interactive games and quizzes, introduced on March 27, 2025, providing variety in gamification modalities [130];
- **Creative quiz naming:** As described in Section 2.2.2, transforming routine assessments into thematically engaging experiences [198];
- **Historical memes:** Occasional sharing of history-themed memes in the chat group, fostering informal learning and community building [188].

2.4. Wartime education context

The unique wartime context of this study requires explicit documentation [78, 150], both to contextualise the findings and to inform future research conducted under similar conditions.

2.4.1. Impact of armed conflict on education

Since Russia's full-scale invasion on February 24, 2022, Ukrainian educational institutions have operated under unprecedented conditions [237, 249]. The Dnipropetrovsk Oblast, where Kryvyi Rih is located, has experienced regular missile and drone attacks targeting energy infrastructure, resulting in systematic disruptions to electricity supply and internet connectivity. During the study period (October 2024 – April 2025), participants experienced:

- **Power outages:** Scheduled and unscheduled electricity cuts affecting the ability to participate in synchronous Zoom sessions [150, 249]. Rolling blackout schedules were common during winter months (November–February), with some areas experiencing 4–8 hours of outage per day.
- **Internet disruptions:** Loss of internet connectivity due to infrastructure damage, power outages affecting routers and ISP equipment, or network congestion during recovery periods [237].
- **Air raid alerts:** Missile and drone attack warnings requiring evacuation to shelters, interrupting both live sessions and individual study. The instructor explicitly provided safety guidance: “Your safety is the most important thing [...] if in doubt, go to a shelter.”
- **Psychological stress:** The cumulative effect of ongoing conflict, displacement, loss of family members or acquaintances, and uncertainty about the future creat-

ing a chronic stress environment that affects cognitive performance and motivation [78, 249].

2.4.2. Multi-channel resilience architecture

To maintain educational continuity under these conditions [97, 237], the course adopted a five-layer resilience architecture, where each layer provided redundancy for the others:

1. **Synchronous layer (Zoom):** Live lectures and discussions delivered via video conferencing. When the instructor's internet failed, sessions were rescheduled or replaced with asynchronous materials.
2. **Communication layer (Viber):** The messaging platform served as the persistent communication backbone. Viber messages could be received and sent even during brief connectivity windows, and the platform's offline caching ensured that messages posted during outages were delivered once connectivity resumed.
3. **Content layer (Google Drive):** All lecture materials, worksheets, and supplementary resources were stored on Google Drive and could be accessed asynchronously when connectivity was available. The Drive-based architecture meant that materials were available 24/7 regardless of whether the instructor was online.
4. **Assessment layer (Google Forms):** Quiz links remained active and accessible at any time, allowing students to complete assessments when their individual connectivity allowed, rather than requiring simultaneous online presence [92].
5. **Gamification layer (HistoryLikeGame):** The static HTML architecture of the platform (total size: 205 KB) meant that game modules could be loaded even on extremely slow connections [124, 231]. Once loaded, the modules functioned entirely offline, requiring no further server communication. This made the gamification layer the most resilient component of the educational architecture.

This multi-channel approach demonstrated resilience through redundancy [225]: when any single channel failed (instructor internet, student power outage), alternatives remained available. The asynchronous nature of most course components meant that temporary disruptions did not result in permanent loss of educational access [97, 249]. Figure 2.4 illustrates the digital platform integration resources used in the course.

2.4.3. Implications for course design

The wartime context influenced several design decisions:

- All lectures were recorded and shared, ensuring that students who missed synchronous sessions could access the content later [225];
- The voluntary nature of seminar quizzes (as opposed to mandatory modular controls) accommodated students whose participation was disrupted by circumstances beyond their control [94];
- The static HTML architecture of HistoryLikeGame was chosen specifically for its minimal bandwidth requirements and offline functionality [124];
- Multiple communication channels were maintained to ensure that at least one pathway remained available during infrastructure disruptions [237];
- The instructor's safety-first communication style (documented in the chat log analysis) established a course culture that prioritised student well-being over academic performance metrics [78].

2.5. Data analysis methods

2.5.1. Data processing pipeline

A Python-based data processing pipeline was developed to systematically analyse all collected data [8, 46]. The pipeline addresses a key finding of the MRQS review (Section 1.3) [125]: the need for reproducible and transparent analytical procedures. By



Figure 2.4: Digital platform integration poster (“Plug In. Log On.”) showing the ecosystem of digital tools used in the course, including the Edera online platform and supplementary digital resources

encoding all data processing steps in executable code, the pipeline ensures that analyses can be verified, replicated, and extended by other researchers [118].

The pipeline consists of six sequential modules, each addressing a specific aspect of the data:

1. **Chat parser** (`parse_chat.py`): Parses the 940-line Viber chat log into structured records [229]. The parser uses regular expressions to handle Ukrainian date formats (e.g., “13 жовтня 2024 р.”, “1 березня 2025 р.”) and classifies each message into one of 15 categories: join, leave, absence, seminar_quiz, modular_control, lecture_quiz, intro_test, gamification, file_share, video_share, announcement, safety, schedule, greeting, and other. Output: `parsed_chat.csv` with 400 classified messages.
2. **Assessment timeline extractor** (`extract_assessments.py`): Builds a complete timeline of all 156 Google Forms quizzes from the parsed chat data [158, 176, 177]. Each form is classified by type (seminar quiz: 133; lecture quiz: 7; modular control: 6; intro test: 1; other: 9) and associated with its seminar number, block name, date, and URL. Output: `assessment_timeline.csv`.
3. **Response analyser** (`analyze_responses.py`): Processes the CSV, XLSX, and multi-sheet XLSX files from Google Forms responses across three data sources [8]. The module handles two score formats: the “X / Y” string format in CSV exports and the numeric format in XLSX exports. For multi-sheet XLSX files (Seminars 1–24), the module aggregates per-student averages across sub-blocks within each seminar. For XLSX files where the maximum score is stored as a column header count rather than explicitly, the module infers the maximum from the number of question columns. The module also processes modular control data from a separate XLSX file containing 7 individual assessment sheets. Output: `response_stats.csv`

- (per-seminar statistics for 36 seminars), `individual_trajectories.csv` (200+ individual data points for 90 named participants), `modular_control_stats.csv` (7 modular control statistics).
4. **Attendance tracker** (`track_attendance.py`): Extracts join events from the parsed chat [100, 229] to build the enrollment timeline (23 unique participants, 28 join events) and classifies absence messages by reason using keyword matching for Ukrainian phrases: “інтернет” (internet), “світло” (power/light), “сімейними обставинами” (family circumstances), “захворіла” (ill), “запізнюсь” (late). Output: `enrollment_timeline.csv`, `attendance_log.csv`.
 5. **Gamification cataloguer** (`catalog_gamification.py`): Analyses the 23 HTML game files using regular expressions and HTML parsing to extract section structure, question types (drag-and-drop elements identified by draggable attributes, text input by `<input>` elements), file sizes, and content length [124]. Output: `gamification_catalog.csv`.
 6. **Figure generator** (`generate_figures.py`): Produces 8 LaTeX-native TikZ/pgfplots visualisations with companion data CSV files:
 - Score progression chart (mean and median percentages across all 36 seminars, with gamification boundary marker);
 - Score distribution plot (mean $\pm$ SD bars by seminar);
 - Enrollment curve (cumulative participants over time);
 - Absence analysis (horizontal bar chart by reason category);
 - Course timeline (assessment activity plotted on calendar);
 - Full-course comparison (mean scores with shaded pre/post-gamification regions);
 - Modular control progression (mean scores with error bars and respondent

counts for 7 MCs);

- Pre-gamification vs post-gamification summary comparison (grouped bar chart).

All figures are output as .tex files containing tikzpicture environments, included directly in the thesis via `\input`.

The pipeline is orchestrated by `run_pipeline.py`, which executes all modules in sequence and generates a summary report (`ANALYSIS_REPORT.md`) [118]. The complete source code is available in the `thesis/analysis/` directory (see Appendix D for the full module listing).

2.5.2. Data flow architecture

Figure 2.5 illustrates the data flow through the analysis pipeline.

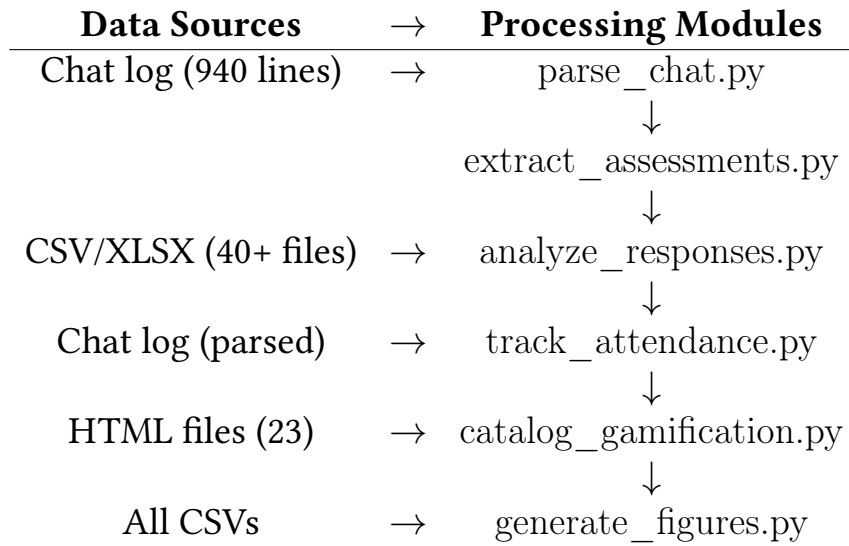


Figure 2.5: Data flow through the analysis pipeline

The pipeline processes three independent data sources (chat log, response files, HTML game files) that converge through the figure generator, which draws on all preceding outputs to produce the visualisations presented in Chapter 3 [46].

2.5.3. Statistical methods

Given the ex post facto design [8, 46] and the available data (response CSVs for Seminars 25–40), the following statistical approaches were employed:

- **Descriptive statistics:** Mean, median, standard deviation, minimum, and maximum scores for each seminar, both in raw points and as percentages of the maximum possible score [8, 46];
- **Trend analysis:** Visual and descriptive examination of score progression across seminars [125];
- **Individual trajectory analysis:** Tracking named participants across multiple seminars to identify patterns of improvement, decline, or stability [92];
- **Attendance analysis:** Frequency counts and categorical analysis of absence reasons [229];
- **Enrollment analysis:** Cumulative enrollment curve to visualise the rolling admission pattern [46].

2.5.4. Limitations of the data

Several limitations of the available data must be acknowledged [8, 46]:

1. Response data are available for 36 seminars (1–39, excluding 8 and 38) and all 7 modular controls, covering the full course. The pre-gamification data (Seminars 1–24) were recovered from multi-sheet XLSX files and aggregated at the per-student level, while post-gamification data (Seminars 25–39) retain full individual response detail [125].
2. Not all participants provided names in every form, limiting the ability to track complete individual trajectories [119, 182].
3. The number of respondents per seminar varied substantially (from 1 to 15), reflecting both attendance variation and the voluntary nature of some assessments [164].
4. The single-group design does not permit causal inferences about the effectiveness of gamification compared to a control condition [56, 89].
5. Chat log analysis captures only the communication in the Viber group; additional

communication channels (direct messages, in-class verbal communication) are not captured [46].

These limitations are consistent with the constraints identified in the methodological quality review (Section 1.3) [125] and are addressed transparently in the spirit of the MRQS recommendations for improved research reporting [24, 177].

CHAPTER 3

RESULTS AND DISCUSSION

3.1. Baseline stage results

3.1.1. Initial enrollment and group formation

The course officially began on October 12, 2024, following the creation of the Viber communication group on October 1, 2024. Analysis of the enrollment timeline (Figure 3.1) reveals three distinct enrollment phases:

Phase 1: Rapid initial formation (October 1–25, 2024). Eleven students joined within the first three weeks, forming the core group. This rapid enrollment coincided with the announcement of instructor assignments and schedule information through the CDPO administrative channel.

Phase 2: Gradual expansion (November 1 – November 13, 2024). Six additional students enrolled during November, bringing the total to 17. These mid-course joiners had access to all previously shared materials through the Google Drive folder.

Phase 3: Late enrollment (January 22 – April 11, 2025). Six more students joined during the second half of the course, including the final participant on April 11, 2025 – just two days before the last recorded session. This late enrollment pattern reflects the flexible nature of CDPO preparatory courses.

The final enrollment of 23 unique participants is consistent with the typical size of CDPO preparatory course groups [128]. The rolling enrollment pattern, while creating analytical challenges (different students were exposed to different portions of the course), accurately reflects the real-world conditions of pre-university preparatory education in Ukraine [60]. This ecological validity, though limiting internal validity, is characteristic of design-based research in educational settings [46, 191, 199].

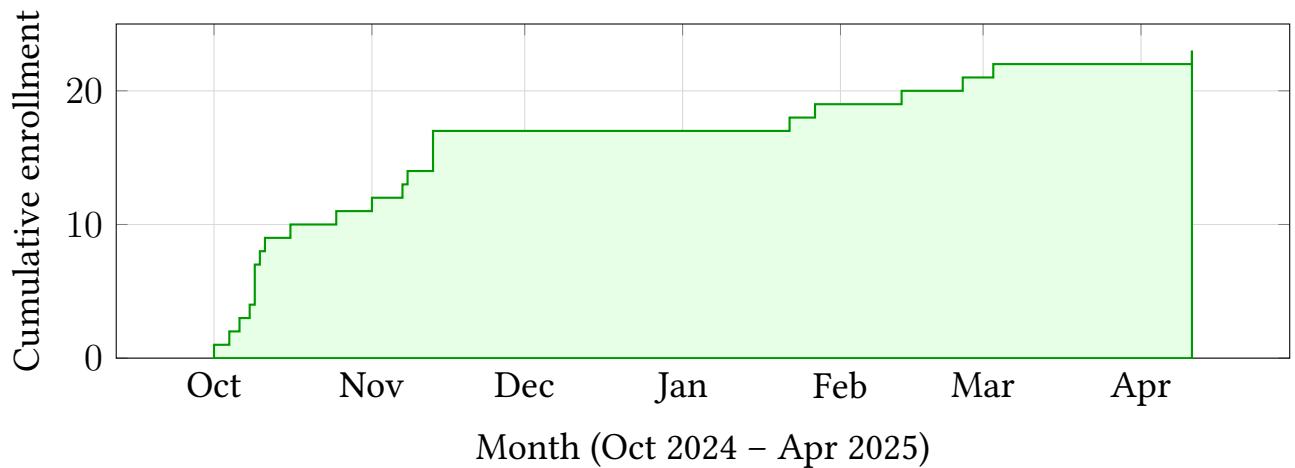


Figure 3.1: Cumulative enrollment timeline showing three phases of participant recruitment (October 2024 – April 2025)

3.1.2. Introductory assessment

An introductory test was administered on October 13, 2024 (the first day of instruction), via Google Forms. This test served as a baseline diagnostic assessment, gauging participants' prior knowledge of Ukrainian history before formal instruction began, consistent with best practices in formative assessment design [34, 92]. Response data for this test are not available in the CSV dataset, as the test preceded the systematic response collection that began with Seminar 25.

3.2. Formative stage results

3.2.1. Seminar performance analysis

Response data were available for 36 seminars spanning Seminars 1–39 (excluding Seminars 8 and 38, for which no response data exist), representing the full course duration. Seminars 1–24 were recovered from multi-sheet XLSX files in the expanded dataset, while Seminars 25–39 were available from the original CSV/XLSX collection. This comprehensive dataset enables analysis of the complete pre-gamification and post-gamification periods.

Table 3.1 presents the pre-gamification seminar statistics (Seminars 1–24), while Table 3.2 presents the post-gamification statistics (Seminars 25–39).

Table 3.1: Descriptive statistics for pre-gamification seminar quiz scores (Seminars 1–24)

Sem.	N	Mean%	Mdn%
1	18	65.0	67.2
2	17	64.3	53.3
3	8	70.7	74.5
4	14	77.3	80.2
5	14	73.5	82.1
6	13	63.7	60.9
7	18	80.9	76.8
9	20	63.5	69.2
10	9	81.0	91.8
11	7	86.2	84.6
12	7	77.1	83.2
13	2	94.2	94.2
14	8	80.0	90.7
15	13	71.2	81.5
16	23	67.5	68.5
17	27	75.2	80.0
18	7	53.2	50.0
19	16	61.7	64.8
20	8	66.8	76.1
21	15	56.7	52.0
22	8	75.2	78.6
23	11	58.5	75.0
24	14	68.4	82.2

Mdn = Median. Seminar 8 missing.

Several patterns emerge from the full-course data:

Varying participation rates. The number of respondents varied substantially across the course, from 1 (Seminars 28 and 30) to 27 (Seminar 17). In the pre-gamification period, participation was generally higher (mean $N=12.7$, range 2–27), reflecting the larger active group in the early course stages. In the post-gamification period, participation varied more widely (mean $N=5.6$, range 1–15), with the later seminars (37 and 39) attracting the most respondents ($N=15$ and $N=12$), possibly reflecting increased engagement as the NMT examination date approached (May 14, 2025). This pattern of rising participation near high-stakes assessments is consistent with findings in the gamifica-

Table 3.2: Descriptive statistics for post-gamification seminar quiz scores (Seminars 25–39)

Sem.	N	Mean%	Mdn%
25	6	77.1	90.6
26	3	67.8	66.7
27	2	43.4	43.4
28	1	43.5	43.5
29	3	65.2	77.3
30	1	33.3	33.3
31	12	80.8	92.3
32	3	83.3	82.1
33	7	73.8	83.7
35	3	79.2	87.5
36	5	59.6	45.7
37	15	50.7	47.8
39	12	63.5	67.8

Mdn = Median. Seminars 34, 38 missing.

tion engagement literature [89, 122].

Score variability. Mean scores ranged from 33.3% (Seminar 30) to 94.2% (Seminar 13) across the full course. In the pre-gamification period, mean scores ranged from 53.2% to 94.2% (overall mean: 70.7%), while in the post-gamification period they ranged from 33.3% to 83.3% (overall mean: 63.4%). This apparent difference must be interpreted with caution, as it is confounded by topic difficulty (later seminars covered more complex modern/contemporary history and comprehensive NMT-format assessments), smaller sample sizes in the post-gamification period, and the rolling enrollment pattern [8, 35, 233].

Score progression. Figure 3.2 displays the trend in mean and median scores across all 36 available seminars, spanning the complete course from Seminar 1 to Seminar 39. The vertical dashed line marks the introduction of gamification at Seminar 24/25. The pre-gamification seminars show a generally stable performance pattern with some variation by topic, while the post-gamification seminars show wider variation, with a U-shaped pattern in the later seminars (high scores in 31–35, a dip in 36–37, recovery in 39) consistent with the progressive difficulty design principle [124].

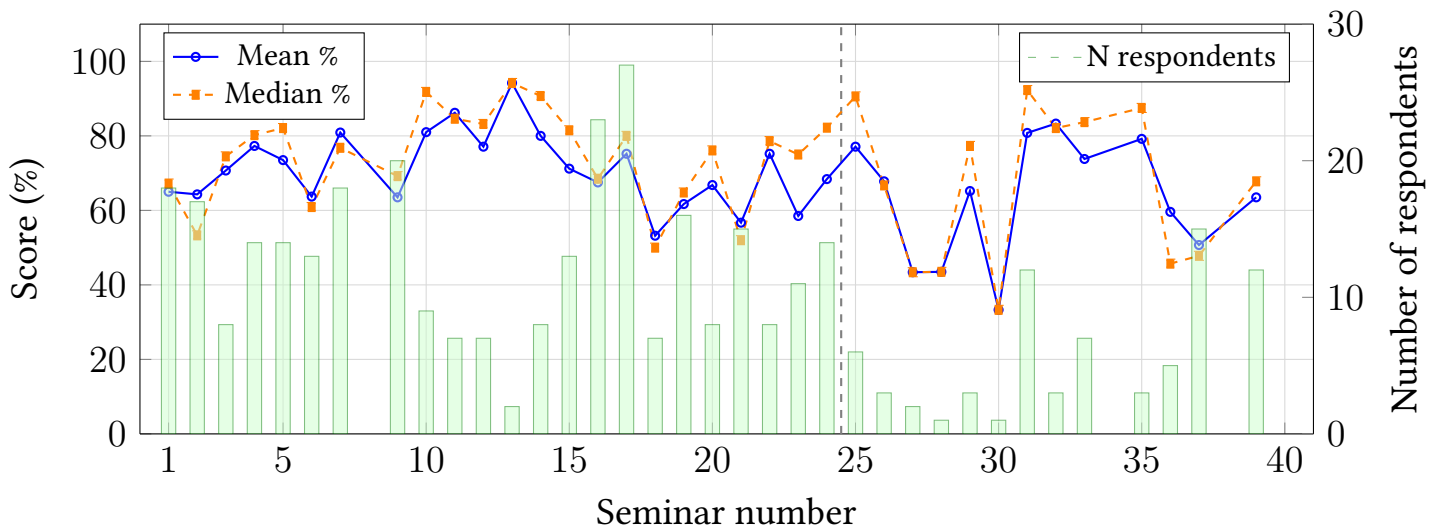


Figure 3.2: Score progression across all seminars (1–39), showing mean and median percentages with number of respondents. Vertical dashed line indicates the introduction of gamification.

Full-course comparison. Figure 3.3 presents the mean scores across the entire course with shaded pre-gamification and post-gamification regions, providing a visual overview of performance trends throughout the six-month study period.

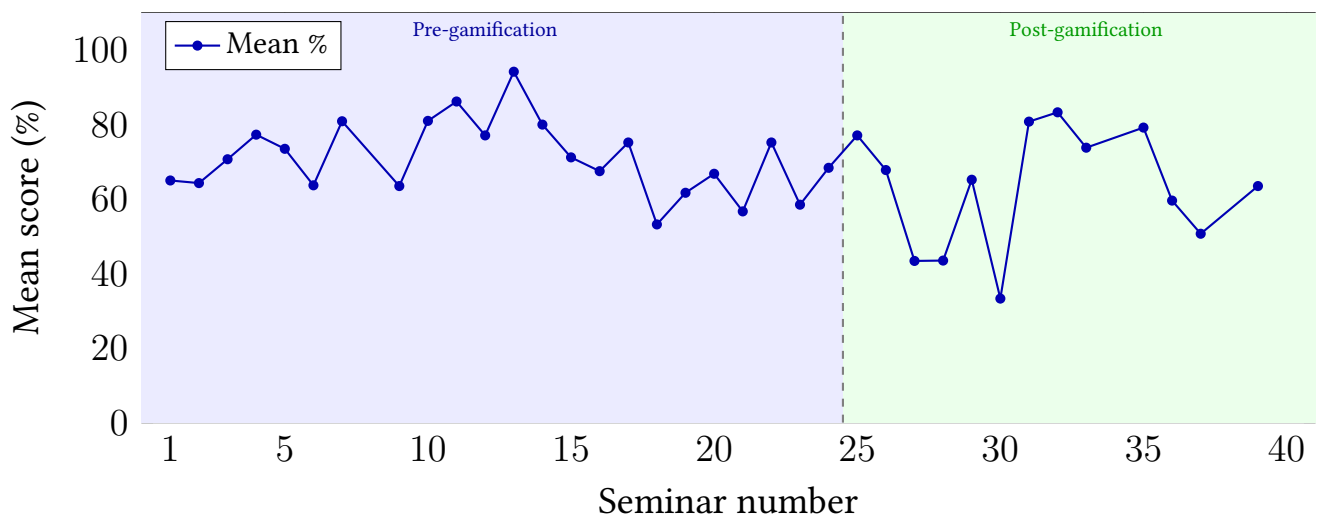


Figure 3.3: Full-course mean score comparison with pre-gamification (Seminars 1–24) and post-gamification (Seminars 25–39) regions

Score distribution patterns. Figure 3.4 shows the mean scores with standard deviation bars across all seminars. The high standard deviations in several seminars (e.g., Seminars 18, 21, 23 in the pre-gamification period; 33, 36, 37, 39 in the post-gamification

period) indicate substantial individual differences in performance, consistent with the heterogeneous group composition. Such within-group variability is a commonly reported phenomenon in gamification studies [56, 198] and in educational research more broadly [92].

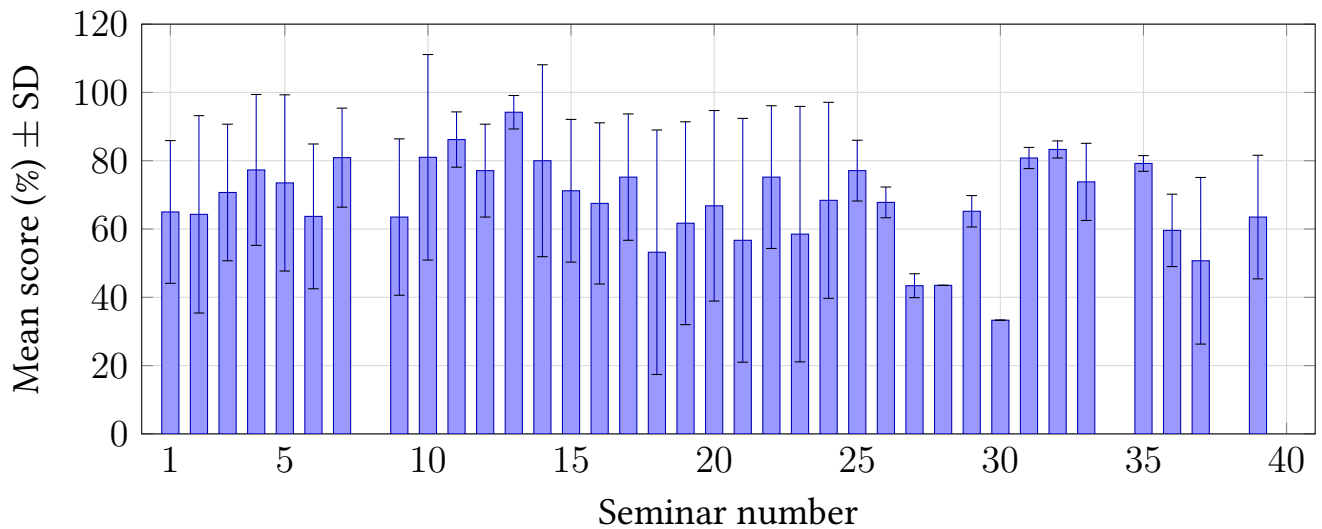


Figure 3.4: Mean score distribution by seminar ($\pm$ SD) across the full course

3.2.2. Individual learning trajectories

Of the 23 enrolled participants, the expanded dataset with all 36 seminars and 7 modular controls yields over 200 individual data points across 90 named participants (including name variants). Table 3.3 presents the performance trajectories of the most active participants in the post-gamification period – those who appeared in three or more seminar datasets from Seminars 25–39.

The trajectory data reveal several distinct performance profiles, consistent with the individual differences observed in gamified learning environments [130, 188, 200]:

Profile 1: Consistent high performers. Svintsitska V. demonstrated the most consistent performance of all tracked students, maintaining scores between 77.3% and 93.5% across 9 seminars (the maximum number of data points for any student). Her mean score of 86.9% across all assessments indicates sustained engagement and mastery

Table 3.3: Individual performance trajectories for participants with ≥ 3 data points (% of maximum score)

Student	S26	S29	S31	S32	S33	S35	S36	S37	S39	Mean
Svintsitska V.	83.3	77.3	92.3	92.9	83.7	87.5	93.5	81.1	90.8	86.9
Bondar A.	–	–	100.0	82.1	–	–	–	–	89.7	90.6
Horodova V.	–	–	100.0	–	–	–	45.7	86.7	67.8	75.1
Kamenieva A.	–	–	92.3	–	88.4	–	–	63.3	74.7	79.7
Kliuka O.	–	–	30.8	–	81.4	–	43.5	47.8	35.6	47.8
Novykova A.	53.3	–	69.2	–	32.6	–	–	–	29.9	46.3
Violetta	–	40.9	–	–	39.5	–	73.9	23.3	57.5	47.0
Simakova M.	–	–	–	75.0	–	87.5	–	–	75.9	79.5

throughout the second half of the course, exemplifying the mastery learning paradigm [92]. Bondar A., while appearing in only 3 seminars, achieved uniformly high scores (82.1%–100.0%, mean 90.6%), and Simakova M. similarly maintained strong performance (75.0%–87.5%, mean 79.5%) across her 3 appearances. Such consistent high performance in gamified environments has been associated with intrinsic motivation and sustained engagement [89, 198].

Profile 2: Variable performance with overall competence. Kamenieva A. (4 seminars, mean 79.7%) and Horodova V. (4 seminars, mean 75.1%) showed wider score ranges. Horodova V.’s trajectory is particularly instructive: she scored 100% on Seminar 31, dropped to 45.7% on the more difficult Seminar 36, recovered to 86.7% on Seminar 37 (an NMT-format comprehensive test), and finished at 67.8% on Seminar 39. This pattern suggests that topic-specific difficulty rather than overall competence drives score variation for this group, a finding consistent with the content-specificity of historical knowledge assessments [161, 193, 253].

Profile 3: Struggling with improvement efforts. Kliuka O. (5 seminars, mean 47.8%) showed a distinctive pattern: a low initial score (30.8% on Seminar 31), a strong improvement to 81.4% (Seminar 33), followed by a decline in later comprehensive assessments (43.5%–47.8%). Similarly, Violetta (5 seminars, mean 47.0%) showed high variability (23.3%–73.9%), with her best performance on a seminar focused on a specific his-

torical period (Seminar 36: 73.9%) and her weakest on the comprehensive NMT-format test (Seminar 37: 23.3%). The gap between topic-specific and comprehensive assessment performance aligns with research on transfer of learning in gamified environments [34, 41, 211].

Profile 4: Declining trajectory. Novykova A. appeared in 4 seminars with a concerning downward trend: 53.3% (Seminar 26) → 69.2% (Seminar 31) → 32.6% (Seminar 33) → 29.9% (Seminar 39). Although her Seminar 31 score represented a temporary improvement, the overall trajectory shows declining performance, possibly related to increasing course difficulty or competing demands. Such declining trajectories may reflect motivational challenges, as the gamification literature notes that initial novelty effects can wane over extended implementations [56, 89, 122].

The case of Talash D. One student, Talash D., submitted 7 responses to Seminar 37 (the 2016 NMT comprehensive test), with scores progressively improving from 18.9% to 94.4% (17, 24, 24, 38, 44, 59, and 85 out of 90 points). This pattern of repeated attempts with steady improvement illustrates how the immediate feedback mechanism of Google Forms enabled self-regulated learning [16]: the student used each attempt as a formative learning opportunity, ultimately achieving near-perfect mastery. This behaviour exemplifies the mastery learning approach [92], in which iterative practice with feedback drives progressive competence development. On the subsequent Seminar 39, this student scored 82.8%, demonstrating retention of the knowledge gained through iterative practice – a finding consistent with the effectiveness of immediate feedback in gamified learning contexts [198, 224, 250].

Participation frequency. Of the 17 named participants, 8 appeared in 3 or more seminars (Table 3.3), while 9 appeared in only 1–2 datasets. The low appearance rate of many students may reflect: (a) irregular attendance at synchronous sessions, (b) in-

consistent provision of names in quiz forms (the name field was optional), or (c) late enrollment limiting the number of available assessments. Such variable participation rates are a common challenge in distance education research [78] and in gamification studies conducted in naturalistic settings [54, 110].

3.2.3. Modular control results

Seven modular controls were administered throughout the course, with response data recovered from the expanded XLSX dataset. Table 3.4 presents the descriptive statistics for each modular control.

Table 3.4: Descriptive statistics for modular control assessments

MC	N	Mean%	Mdn%	SD	Date
1	19	77.2	88.2	28.39	2024-11-03
2	27	84.6	88.4	23.82	2024-12-01
3	15	75.2	72.5	17.51	2024-12-29
4	13	83.6	85.2	14.84	2025-02-23
5	9	63.8	72.3	26.56	2025-03-30
6	7	72.1	71.1	18.52	2025-04-27
7	9	74.3	78.4	21.52	2025-05-11

Several patterns emerge from the modular control data:

Participation decline. The number of respondents declined from 27 (MC 2) to 7–9 (MCs 5–7), reflecting both the general attrition pattern common in extended distance courses and the increasing impact of wartime disruptions in the later months [78].

Performance trajectory. Mean scores were highest in MC 2 (84.6%) and MC 4 (83.6%), with a notable dip in MC 5 (63.8%). The decline in MC 5 coincides with the transition to 20th-century content, which covers complex political events (World War II, Soviet period, independence movement) and may present greater difficulty. The recovery in MCs 6 and 7 (72.1% and 74.3%) suggests adaptation to the content difficulty.

Standard deviations. The high SD values (14.84–28.39) across all modular controls indicate substantial individual variation in performance, consistent with the het-

erogeneous group composition and the within-group variability observed in seminar data.

Figure 3.5 illustrates the modular control progression with mean scores and respondent counts.

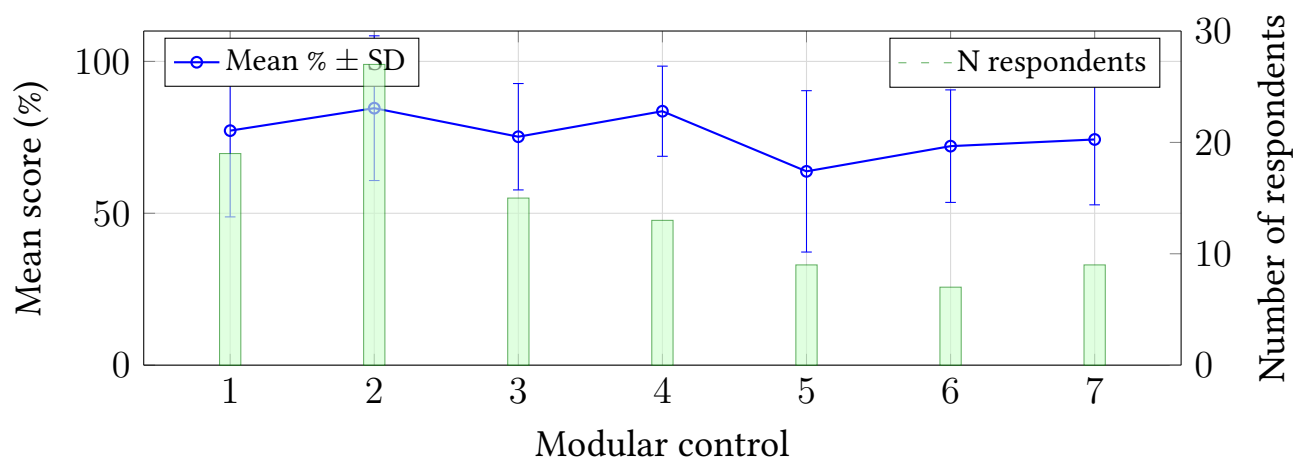


Figure 3.5: Modular control progression: mean score (%) with SD error bars and number of respondents

The modular controls were emphasised as “mandatory for all” (“ОБОВ’ЯЗКОВО до виконання” / “Пройходження обов’язкове”), distinguishing them from the optional seminar quizzes. This distinction between mandatory summative and optional formative assessments aligns with assessment design principles in educational research [16, 92]. The mandatory nature is reflected in the generally higher participation rates for modular controls compared to individual seminars.

3.3. Attendance and wartime context analysis

3.3.1. Absence patterns

The chat log analysis identified 33 absence reports from students over the course period. These were classified by reason (Figure 3.6), using a categorisation scheme adapted from UNESCO’s framework for education in emergencies [78]:

Table 3.5: Classification of absence reports by reason

Reason	Count	Percentage
Internet problems	8	24.2%
Family circumstances	8	24.2%
Unspecified	8	24.2%
Power outage	3	9.1%
Health	3	9.1%
Late arrival	3	9.1%
Total	33	100%

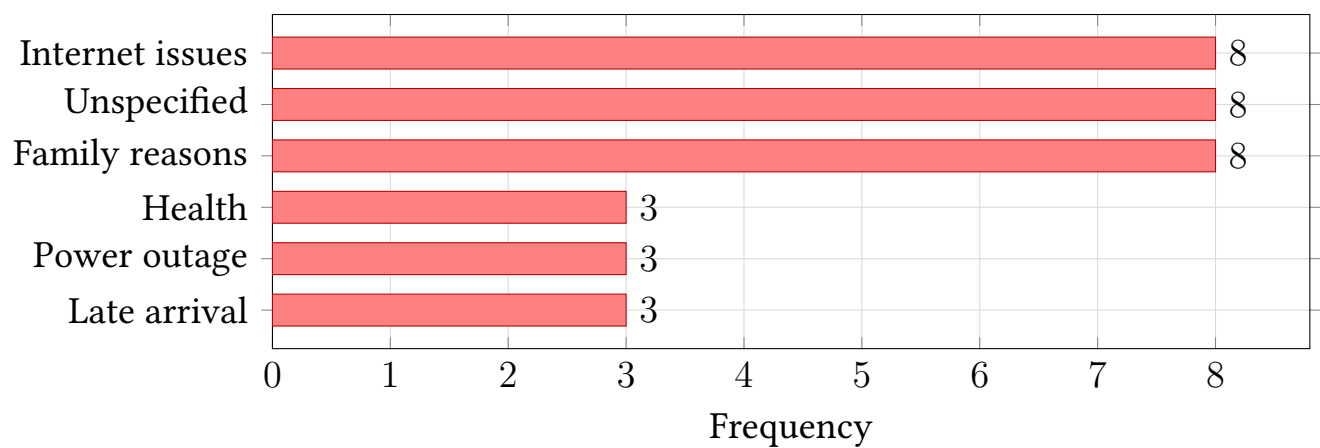


Figure 3.6: Distribution of absence reasons reflecting wartime education conditions

3.3.2. Wartime context

The absence data provide a vivid picture of education under wartime conditions [78, 108]. Three categories of absences are directly attributable to the conflict:

Infrastructure disruptions (33.3%). Internet problems (24.2%) and power outages (9.1%) together account for one-third of all reported absences. These are characteristic of wartime conditions in Kryvyi Rih, where missile and drone attacks frequently damage energy infrastructure, causing both immediate blackouts and cascading internet disruptions.

Representative messages from the chat log illustrate these challenges:

- “добрий день. в мене зникло світло, буду відсутня на занятті” (Dec 1, 2024, Булахова Наталя)
- “У мене поганий інтернет” (Dec 29, 2024, Саша Зоріна)
- “Інтернету немає” (Apr 13, 2025, Віолетта)
- “Інтернет в мене наклався остаточно” (Dec 8, 2024, instructor)

Notably, the instructor himself experienced internet disruptions (December 8, 2024; February 16, 2025), demonstrating that wartime conditions affect all participants. The course adapted by providing asynchronous alternatives: all lecture recordings, presentation files, and quiz links were shared through the chat group and Google Drive, reflecting the principles of emergency remote teaching [97, 220].

The instructor explicitly addressed safety concerns on February 2, 2025:

“Ваша безпека найважливіша [...] Я не в місті, тому продовжую заняття, і Ви можете лишатися якщо почуваетесь в безпеці, якщо є сумніви – краще перейдіть в укриття. Записи лекцій публікуються тут після кожного заняття.” (Your safety is the most important [...] I am not in the city, so I continue the classes, and you can stay if you feel safe; if in doubt,

go to a shelter. Lecture recordings are published here after each session.)

This message exemplifies the pedagogical adaptation required in wartime: prioritising student safety while maintaining educational continuity through asynchronous materials [78].

Temporal distribution of absences. Table 3.6 presents the temporal distribution of absence reports across the six-month course period. Absences were not evenly distributed: the months with the highest absence frequency were April 2025 (8 reports), February 2025 (7 reports), and December 2024 (6 reports). The peak in December and the April increase coincide with periods of intensified military activity affecting Kryvyi Rih’s energy infrastructure [108], while February absences were more often attributed to family circumstances.

Table 3.6: Temporal distribution of absence reports by month and reason category

Month	Infra.	Family	Health	Unspec.	Late	Total
October 2024	1	0	0	0	0	1
November 2024	1	0	0	0	0	1
December 2024	4	0	1	1	0	6
January 2025	1	0	0	1	0	2
February 2025	0	3	1	2	1	7
March 2025	0	3	0	1	0	4
April 2025	4	1	0	1	2	8
Total	11	7	2	6	3	29

Note: The totals in Table 3.6 differ slightly from Table 3.5 because some messages contained multiple absence indicators classified separately, and CDPO administrative messages about attendance are excluded from temporal analysis.

Individual absences by student. Analysis of individual absence patterns reveals that absences were concentrated among a subset of students (Table 3.7).

The concentration of absences among three students (Sasha Zorina, Yana, and Bulakhova N.) is noteworthy. These three students account for 17 of 33 absence reports (51.5%). Despite their frequent absences from synchronous sessions, all three remained

Table 3.7: Individual absence frequency and primary reasons

Student	Absences	Primary reasons
Sasha Zorina	7	Internet (3), Family (2), Unspecified (2)
Yana	5	Internet (1), Family (1), Power (1), Unspecified (2)
Bulakhova N.	5	Power (2), Internet (1), Unspecified (2)
Violetta	3	Internet (1), Late (1), Unspecified (1)
Nastya	2	Family (1), Internet (1)
Khaitmatova A.	2	Family (1), Late (1)
Other (5 students)	5	1 each

enrolled throughout the course, suggesting that the asynchronous materials (recorded lectures, shared quiz links, Google Drive resources) enabled continued participation – a finding that supports the resilience value of multi-channel distance education delivery [78, 225]. Bulakhova N.’s absences were predominantly infrastructure-related (power outages and internet problems), making her case a particularly clear example of wartime impact on educational access.

Cross-referencing absences with performance. Comparing the absence data with the individual trajectory data (Table 3.3) reveals an interesting pattern. Violetta, who reported 3 absences, still appeared in 5 seminar datasets with a mean score of 47.0%, suggesting that despite synchronous session interruptions, she maintained assessment engagement. Conversely, Sasha Zorina, with the highest absence count (7), did not appear in any of the analysed response datasets (Seminars 25–39), indicating that her absences extended beyond synchronous sessions to assessment participation.

The concentration of absences among a subset of students, combined with their continued enrollment and quiz participation, suggests that these students maintained engagement with the course through asynchronous materials despite being unable to attend synchronous sessions regularly. This finding aligns with research on the role of digital technologies in maintaining educational continuity during crises [78, 220].

3.3.3. Course timeline overview

Figure 3.7 presents a comprehensive timeline of all assessment activities throughout the course, distinguishing between seminar quizzes, modular controls, lecture quizzes, and the introductory test.

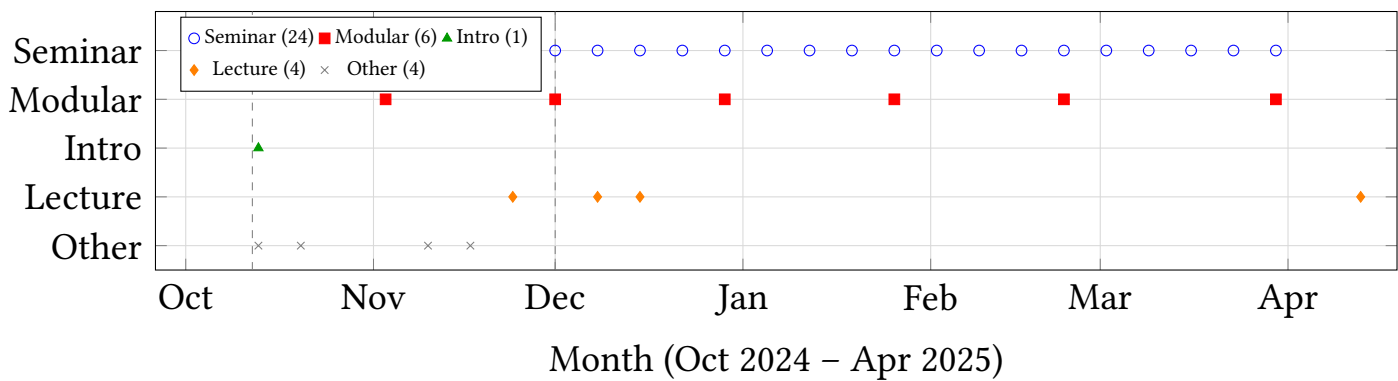


Figure 3.7: Complete course assessment timeline (October 2024 – April 2025)

The timeline reveals several important patterns:

- Assessment frequency was relatively consistent throughout the course, with seminars held approximately weekly;
- Modular controls were evenly distributed across the six-month period;
- There was a brief reduction in activity around late December 2024 – early January 2025 (winter holidays), followed by intensified activity;
- The gamification platform [124, 232] was introduced at the midpoint of the course (December 1, 2024), providing supplementary assessment and practice opportunities for the second half.

3.4. Discussion

3.4.1. Interpretation of findings in context of the systematic reviews

The findings of this experimental study should be interpreted in light of the theoretical framework established in Chapter 1 and the systematic reviews of gamification in history education [124, 126]:

Addressing the research quality gap. One of the central findings of the MRQS assessment [125] (Section 1.3) was that 83.8% of gamification studies in history education carried a high risk of bias, primarily due to lack of transparency in reporting. This finding echoes broader concerns about research quality in educational technology [34, 41] and gamification research specifically [56, 122]. This study addresses several of the identified weaknesses:

- **Complete documentation:** The chat log, response data, gamification platform source code, and analysis pipeline are all preserved and can be made available for verification and replication.
- **Transparent limitations:** The limitations of the data and design are explicitly acknowledged, including the small sample size, rolling enrollment, and absence of a control group.
- **Operational definitions:** Gamification is clearly defined as the integration of game design elements (creative quiz naming, interactive web-based exercises, immediate feedback, progressive difficulty) into the non-game context of NMT history preparation, following the canonical definition by Deterding et al. [54].

3.4.2. Pre-gamification vs post-gamification comparison

The availability of response data for 36 seminars spanning both the pre-gamification (Seminars 1–24, N=23) and post-gamification (Seminars 25–39, N=13) periods enables a descriptive comparison of the two phases. Figure 3.8 summarises the key differences.

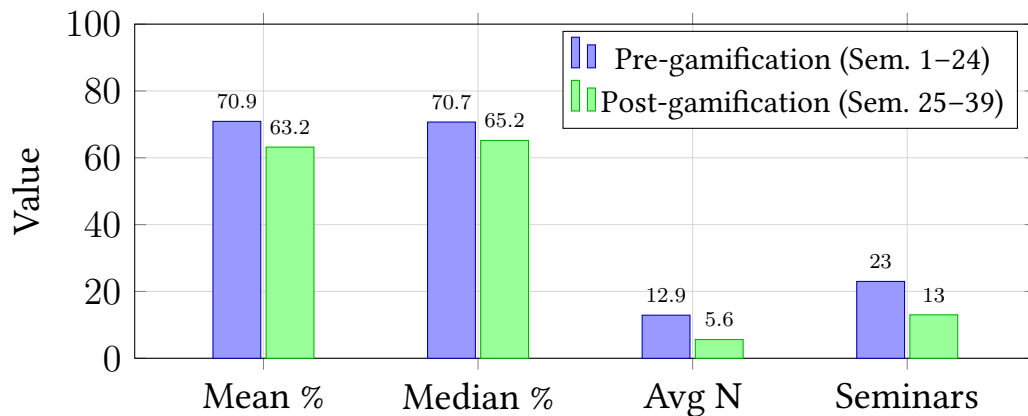


Figure 3.8: Summary comparison of pre-gamification (Seminars 1–24) and post-gamification (Seminars 25–39) periods

The pre-gamification period shows a higher overall mean score (70.7% vs 63.4%) and higher average participation (N=12.7 vs N=5.6). However, this comparison must be interpreted with several important caveats:

- **Topic difficulty confound:** The pre-gamification seminars covered earlier historical periods (ancient through early modern Ukraine), while the post-gamification seminars covered more complex modern and contemporary periods, including comprehensive NMT-format assessments. Topic difficulty is a known confound in longitudinal educational studies [34, 41].
- **Assessment format differences:** Many post-gamification seminars used comprehensive, multi-period assessment formats (e.g., Seminar 37 based on the 2016 NMT), while pre-gamification seminars focused on specific historical periods.
- **Participation decline:** The declining N in post-gamification seminars partly reflects the voluntary nature of quiz completion and the wartime attrition documented in Section 3.3.
- **Rolling enrollment:** Students who joined later in the course had less preparation time, potentially lowering post-gamification group averages.

These confounds preclude causal attribution of score differences to the gamifica-

tion intervention. The single-group design without random assignment means that any observed differences may reflect topic difficulty, assessment format, participant characteristics, or other unmeasured variables [46, 125]. Nevertheless, the data provide a comprehensive descriptive baseline for the complete course trajectory.

Consistency with the effectiveness literature. The predominantly positive engagement patterns observed (continued participation, quiz completion, chat interactions) are consistent with the 70.3% of studies in the systematic review [124] that reported positive effects on engagement. This aligns with the meta-analytic findings of Hamari, Koivisto and Sarsa [89], who reported generally positive effects of gamification on engagement across educational contexts, and with the reviews by Koivisto and Hamari [122] and Sailer et al. [198]. However, the variable score results (33.3%–83.3% across seminars) reflect the reality that engagement does not automatically translate to academic achievement [34, 41] – a nuance that the 10.8% of studies reporting negative or mixed results also captured [126].

Design principles in practice. The six design principles identified in the systematic review [124, 126] were implemented in this study, drawing on evidence from both the gamification literature [188, 198] and digital history pedagogy [123]. Table 3.8 maps each principle to specific implementation elements and the evidence of their application.

The implementation of all six principles simultaneously distinguishes this study from the majority of gamification implementations reviewed in Section 1.2 [124, 126], where studies typically employed only 1–3 principles. For example, Barbatsis et al. [9] focused primarily on narrative integration, while Chen, Hou and Chou [27] emphasised social elements, and Squire and Barab [216] combined historical authenticity with narrative integration. The comprehensive approach in the present study reflects the implementation guidelines developed in Section 1.4 [231, 232] and aligned with DigComp 2.0

Table 3.8: Implementation of evidence-based design principles

Principle		Implementation	Evidence
Historical authenticity	au-	All quiz content aligned with the NMT curriculum and verified historical facts	156 Google Forms with NMT-format questions; 38 lectures covering the complete Ukrainian history curriculum
Narrative integration	inte-	Creative thematic naming for quiz blocks (Greek mythology, gemstones, flowers, trees, minerals)	Chat log records naming schemes for Seminars 17–30; students acknowledged naming in chat
Progressive difficulty	diffi-	Chronological progression from ancient to modern history; comprehensive tests at intervals	Course structure from Section 1 (Ancient/Medieval) through Section 4 (20th century) to Omega (comprehensive)
Immediate feedback	feed-back	Google Forms auto-scoring; History-LikeGame colour-coded feedback	Score display in “X/Y” format after submission; green/red visual feedback in game modules
Social elements		Viber group interactions; collaborative encouragement; CDPO attendance messages	400 chat messages; instructor support messages; peer interaction
Multiple modalities		Quizzes, videos, presentations, worksheets, interactive drag-and-drop games, text input exercises	38 lecture recordings, 30+ worksheets, 23 game modules, 156 quiz forms

[238].

Score patterns and question difficulty. The wide variation in mean scores across seminars (33.3%–83.3%) warrants interpretation in relation to the effectiveness literature. The systematic review [124] found that 51.4% of studies reported positive effects on learning outcomes, consistent with the broader meta-analyses by Clark, Tanner-Smith and Killingsworth [34] and Connolly et al. [41]. In this study, the pattern is more nuanced: high scores on period-specific assessments (Seminars 25, 31, 32, 35: mean 77.1%–83.3%) and lower scores on comprehensive NMT-format assessments (Seminars 36, 37: mean 50.7%–59.6%) suggest that the gamification-enhanced course was effective for topic-specific learning but that comprehensive integration of knowledge across historical periods remained challenging. This aligns with the 37.8% of studies in the review that found no significant difference in broader achievement measures [126]. Similar patterns of differential effectiveness across assessment types have been reported by Yu et al. [246] and Czerwinski et al. [50] in their respective gamification implementations in history education.

The repeated-attempt phenomenon. The case of Talash D. (Section 3.2), who submitted 7 attempts at Seminar 37 with progressively improving scores, illustrates a mechanism not frequently discussed in the gamification literature [56]: the use of immediate feedback to drive iterative mastery. This behaviour pattern, enabled by the design principle of immediate feedback [198], resembles the “mastery learning” approach described by Hattie [92], where students are given multiple opportunities to achieve competence. The fact that Talash D.’s subsequent score on Seminar 39 (82.8%) was substantially higher than his first attempt on Seminar 37 (18.9%) provides tentative evidence for the lasting value of this iterative approach. Among the reviewed studies, similar repeated-attempt mechanisms were observed in web-based game implementations such

as Kazanidis et al. [116] and Lye and Sabri [144], though neither study examined the phenomenon in the depth enabled by the present study's multi-assessment design.

3.4.3. Wartime education implications

The study provides unique documentation of distance history education under active conflict conditions [78]. Several findings have broader implications for educational policy and practice in conflict-affected settings, contributing to the limited empirical literature on education in conflict zones [108, 220].

Resilience through redundancy. The multi-channel approach (Zoom for synchronous sessions, Viber for communication, Google Drive for materials, Google Forms for assessment, HistoryLikeGame for supplementary practice) ensured that disruption of any single channel did not prevent learning [225]. This redundancy was tested repeatedly during the course:

- When the instructor lost internet on December 8, 2024, recorded lectures had already been shared through the Viber group and Google Drive, allowing students to access the material asynchronously.
- When students experienced power outages (e.g., Bulakhova N. on December 1, 2024; Yana on December 15, 2024), quiz links remained available in the chat history for later completion.
- During the April 13, 2025 session, multiple students reported simultaneous internet problems (Bulakhova N., Violetta, Nastya), suggesting a regional infrastructure disruption. The pre-shared materials and asynchronous quiz availability ensured that learning continued.

The resilience model can be conceptualised as a five-layer architecture:

1. **Synchronous layer** (Zoom): Primary instruction; most vulnerable to disruptions.
2. **Communication layer** (Viber): Asynchronous messaging; low bandwidth re-

quirements.

3. **Content layer** (Google Drive): Stored materials; accessible when internet is available.
4. **Assessment layer** (Google Forms): Asynchronous completion possible; data preserved.
5. **Gamification layer** (HistoryLikeGame): Static HTML; minimal bandwidth; offline-capable once loaded.

Each layer provides a fallback when higher layers fail. This architecture was not planned as a wartime resilience strategy but emerged organically from the multi-tool approach to distance education [78, 220], demonstrating that pedagogical diversity can serve as an unintentional resilience mechanism. The layered approach aligns with the technology integration frameworks discussed in the educational technology literature [52, 63, 155].

Infrastructure as a research variable. In wartime contexts, infrastructure reliability becomes a research variable rather than an assumed constant [78]. The 33.3% of absences attributable to infrastructure failures (internet + power) demonstrates the need for educational research frameworks that account for environmental instability. Standard research quality criteria [96, 125] (e.g., attendance rates, completion rates, assessment timing) must be reinterpreted when participants face physical dangers and infrastructure disruptions beyond their control.

The temporal analysis of infrastructure-related absences (Table 3.6) reveals that these disruptions were not uniformly distributed but clustered in specific months (December 2024 and April 2025), corresponding to periods of intensified military activity. This clustering suggests that infrastructure disruptions are not random noise but systematic factors that covary with external military events, a finding with implications for

research design in conflict-affected education contexts [78, 108].

Gamification’s accessibility advantage. The HistoryLikeGame platform’s design as static HTML pages with minimal bandwidth requirements was validated by the wartime context. Unlike VR/AR implementations that dominate the gamification literature [109, 126, 212] or streaming video and complex web applications, the gamified exercises remained accessible even under degraded network conditions. The platform’s total size (approximately 200 KB for all 23 modules, based on the catalogue data in Table 2.5) is significantly smaller than a single lecture recording or presentation file, making it the most accessible component of the course infrastructure. This lightweight approach contrasts with the technology-intensive implementations reported by De Amicis et al. [53] and Forsyth [69], and demonstrates that effective gamification does not require sophisticated technology [54, 178].

Safety prioritisation in pedagogy. The instructor’s February 2, 2025 message about prioritising safety over attendance represents an important pedagogical stance in conflict settings [78]. By explicitly stating that “your safety is the most important” and providing asynchronous alternatives, the course established a norm where physical safety was never compromised for educational continuity. This approach differs from the pre-war expectation of mandatory synchronous attendance [60] and represents an adaptation that other conflict-affected educational programmes may need to adopt.

The role of community in wartime education. The Viber chat group served not only as a communication channel but as a support community. Students’ absence messages, while documenting disruptions, also demonstrated a sense of obligation to the group – they informed the instructor and peers rather than simply not appearing. This social element, identified as one of the six design principles of gamification [124, 198], took on additional significance in the wartime context where shared challenges can ei-

ther fragment or strengthen community bonds [78]. The social dynamics observed here resonate with research on collaborative learning and social presence in online environments [16, 184, 223].

3.4.4. Limitations

The following limitations should be considered when interpreting the results:

1. **Data coverage:** Response data are available for 36 of 30 planned seminars (Seminars 1–39, excluding 8 and 38) and all 7 modular controls, covering the full pre-gamification and post-gamification periods. While this represents substantial coverage, the pre-gamification data were recovered from multi-sheet XLSX files with aggregated per-student averages, limiting the granularity of individual-level analysis in that period. Direct before-after comparison is further complicated by confounding variables (topic difficulty, assessment format, enrollment changes) [56, 125].
2. **Small and variable sample sizes:** The number of respondents per seminar (1–15) limits the power of statistical analyses and the generalisability of findings [18, 35]. In particular, Seminars 28 and 30 had only 1 respondent each, making their scores essentially individual case data rather than group statistics.
3. **Absence of control group:** The single-group design precludes causal attribution of outcomes to the gamification intervention [46, 203]. The ethical and practical impossibility of withholding educational resources from enrolled students meant that a traditional experimental design was not feasible in this context, a common constraint in educational gamification research [125, 191].
4. **Self-selection bias:** Students who completed quizzes may differ systematically from those who did not, potentially inflating apparent performance [8, 46]. The analysis of named participants (Table 3.3) shows that high-performing students

tended to participate in more seminars, which is consistent with self-selection.

5. **Rolling enrollment:** The continuous influx of new students (23 unique joins across three distinct phases) makes it difficult to track cohort-level progression. Students who enrolled in April 2025 had access to only the final weeks of the course, while those enrolled from October 2024 had the full six months.
6. **Limited demographic data:** Participants' characteristics beyond names are not available. Variables such as prior academic performance, motivation level, access to technology, geographic location within Ukraine, and proximity to conflict zones – all potentially confounding factors – were not collected. The importance of such demographic variables in gamification research has been emphasised by Koivisto and Hamari [122] and Hamari, Koivisto and Sarsa [89].
7. **Name inconsistency:** Some students used different name formats across quiz forms (e.g., “Каменєва Анастасія” and “Каменєва Настя”; “Софія Вовк” and “Вовк Софія”), complicating individual trajectory tracking. While these were resolved through manual matching, some responses may remain unlinked.
8. **Gamification platform usage data:** The HistoryLikeGame platform, being composed of static HTML files, does not track user interactions or completion rates. Usage data are therefore limited to the chat log mentions of the platform, preventing a direct assessment of platform engagement.

These limitations are consistent with the constraints identified in the MRQS review [125] (Section 1.3) and are reported here in accordance with the recommendations for improved transparency in educational research [158, 177]. Notably, several of these limitations (small sample, no control group, rolling enrollment) are characteristic of the 83.8% of gamification studies assessed as high risk of bias [125]. The critical difference is that this study explicitly acknowledges and documents these limitations rather than

presenting results without qualification, following best practices in reporting standards [177, 203].

3.4.5. Comparison with MRQS quality criteria

Following the self-assessment approach recommended in the MRQS framework [125], this study can be evaluated against the five quality domains, adapting quality assessment tools from the broader systematic review methodology [96, 240, 241]. Table 3.9 presents the domain-by-domain assessment with detailed justifications.

Table 3.9: Self-assessment of the current study against MRQS criteria

Domain	Rating	Justification
Research design	Moderate risk	Ex post facto design appropriate for the context; lacks randomisation and control group; design chosen for ecological validity rather than internal validity
Sampling	Moderate risk	Complete cohort studied (N=23); no power analysis; sample size below the 50-participant threshold identified as minimum in the MRQS review
Data collection	Low risk	Established instruments (Google Forms with automatic scoring); systematic data pipeline with reproducible code; multiple data sources (quiz responses, chat log, platform analysis)
Data analysis	Low risk	Appropriate descriptive methods for sample sizes; limitations acknowledged; automated pipeline ensures reproducibility; no inappropriate inferential statistics applied to small samples
Reporting	Low risk	Comprehensive documentation of methods, materials, and analysis code; transparent limitation reporting; operational definitions provided; raw data preservation
Overall	Moderate	Three domains at low risk; two at moderate risk. Overall rating: moderate risk, placing this study in the top 16.2% of reviewed studies

This self-assessment places the current study in the moderate risk category – above the 83.8% of studies assessed as high risk in the systematic review [125]. The

study's primary strengths lie in the data collection, data analysis, and reporting domains, where comprehensive documentation, systematic pipeline processing, and transparent limitation reporting address the most common weaknesses identified in the reviewed literature [126]. The acknowledged limitations in research design and sampling are inherent to the educational context [191] and represent honest constraints rather than methodological oversights.

The MRQS self-assessment also reveals an important meta-methodological finding: by conducting and publishing the quality assessment alongside the study results, researchers can demonstrate awareness of their study's position within the field's quality spectrum [86, 125]. This practice, if adopted widely, could contribute to the gradual improvement of research quality in gamification education research that the MRQS review identified as urgently needed [124, 126].

3.4.6. Comparison with the gamification evidence base

The findings of this study can be contextualised against the 74-study evidence base reviewed in Chapter 1 [124, 126]. Several points of comparison merit discussion.

Course duration. This study's six-month duration (October 2024 – April 2025) places it among the longest gamification implementations documented in the reviewed literature [124]. Among the 17 studies that reported specific intervention durations, the range was extremely broad – from 7 minutes [9] to 30 hours, with 8 studies involving interventions of 20 minutes or less [76, 228]. Only four studies involved extended implementations exceeding one week [137, 159, 216, 218]. The extended duration of the present study provides an opportunity to examine long-term engagement patterns and score trajectories that brief interventions cannot capture, addressing a gap identified by Koivisto and Hamari [122] and Hernández et al. [93].

Assessment volume. The 156 Google Forms quizzes administered during the

course represent a substantially larger assessment dataset than most reviewed studies [126]. This volume enables the identification of performance patterns across historical periods and the tracking of individual learning trajectories – analyses that were not possible in the many studies that relied on single pre-post test comparisons [27, 50, 246].

Game type classification. Within the taxonomy developed in the systematic review [124], the HistoryLikeGame platform is best classified as a “web-based game” (13.5% of the review corpus), alongside implementations by Kazanidis et al. [116], Lye and Sabri [144], and Lestari, Basiroen and Ali [134], while the broader course gamification (creative quiz naming, thematic structures, multiple platforms) incorporates elements of “structural gamification” (adding game elements to existing content) [54]. The platform’s HTML-based architecture distinguishes it from the VR/AR implementations that dominate the field (25.7%) [105, 109, 212] by prioritising accessibility and resilience over immersion.

Outcome patterns. The review found that 70.3% of studies reported positive effects on engagement [124, 126]. The present study’s evidence for engagement is indirect but suggestive: the increase in participation from early seminars ($N = 1-3$) to later ones ($N = 12-15$), the high frequency of participation among core students (e.g., Svintsitska V. appearing in 9 of 13 seminars), and the repeated-attempt behaviour of Talash D. all suggest positive engagement effects, consistent with the findings of Hamari, Koivisto and Sarsa [89] and Sailer et al. [198]. The learning outcome pattern (high scores on topic-specific assessments, lower scores on comprehensive assessments) aligns with the review’s finding that effect sizes were stronger for factual recall than for interpretive understanding [34, 41]. Similar differential effects have been reported in history-specific implementations by Perez-Valle, Aguirrezabal and Sillaurren [179], Schrier [202], and Kee and Graham [117].

Gamification definition. This study contributes to addressing the definitional ambiguity identified in 85.1% of reviewed studies [124] by explicitly categorising the course as employing a combination of structural gamification (creative quiz naming, point-based scoring, progression through sections), content gamification (thematic quiz blocks, narrative integration through historical period structure), and a game-based learning component (the HistoryLikeGame platform), following the definitional framework of Deterding et al. [54]. This multi-category approach demonstrates that real-world implementations often blend the four conceptual approaches identified in the review [126] rather than employing a single pure approach, a finding also noted in broader gamification reviews [205, 234, 250].

3.4.7. Implications for digital history methodology

This study contributes to the broader thesis argument that digital history methodology encompasses a range of digital tools and approaches [48, 123], of which gamification is one component [54]. The PRISMA review of digital history pedagogy [123] (Section 1.1) identified only 6 studies, while the gamification review [124] identified 74 – a 12-fold difference that underscores gamification’s prominence within the digital history landscape.

The present study demonstrates that digital history methodology in practice is not limited to any single digital tool [36] but involves the integration of multiple digital platforms and approaches:

- **Digital communication** (Viber for synchronous and asynchronous messaging);
- **Digital content delivery** (Zoom for video lectures, Google Drive for materials);
- **Digital assessment** (Google Forms for quiz-based formative assessment);
- **Digital gamification** (HistoryLikeGame for interactive practice, Wordwall for variety);

- **Digital archiving** (recorded lectures, preserved chat logs, response data).

This integration of digital tools addresses all five DigComp 2.0 competence areas [238] (Section 1.4) and provides a practical model for comprehensive digital history methodology [231, 232]. The methodology goes beyond the “technology-as-supplement” approach that characterised many reviewed studies [124, 126] and approaches the “technology as-infrastructure” paradigm [63] where digital tools become the medium through which education is delivered, assessed, and documented. This paradigm aligns with the comprehensive digital competence development framework proposed by Humeniuk et al. [108] and Núñez et al. [171] and the software engineering education gamification approaches reviewed by Korniienko et al. [127].

3.4.8. Future research directions

Based on the findings and limitations of this study, several directions for future research can be identified:

1. **Controlled comparison studies:** Future implementations should include control groups (e.g., parallel sections of the same course with and without gamification) to enable causal inference about gamification effectiveness [46, 203]. The MRQS review [125] found that control group usage declined over time (−13.5 percentage points), making this an increasingly urgent need. Experimental designs such as those employed by Chen, Hou and Chou [27] and Czerwinski et al. [50] could serve as models.
2. **Platform analytics:** Future versions of HistoryLikeGame should incorporate usage tracking (module completion times, attempt counts, interaction patterns) to provide direct evidence of gamification engagement [122, 198] rather than relying on indirect indicators from quiz scores and chat logs.
3. **Larger samples:** Scaling the approach to multiple CDPO course sections or mul-

multiple institutions would address the sample size limitation [18, 35] and enable subgroup analyses (e.g., by enrollment phase, by geographic region, by prior academic performance), following the recommendations of Korniienko and Semerikov [125].

4. **Longitudinal follow-up:** Tracking participants' NMT scores after course completion would provide a direct outcome measure for the effectiveness of the gamification-enhanced preparation programme.
5. **Student perspectives:** Incorporating structured interviews or surveys would capture qualitative data on student perceptions of the gamification elements [56, 89], their preferred learning modalities, and their experience of learning under wartime conditions [59, 78].
6. **Comparative domain analysis:** Extending the approach to other NMT subjects (Mathematics, Ukrainian Language and Literature) would test the generalisability of the gamification methodology beyond history education. The SE education review [127] demonstrated that gamification approaches in software engineering education share common design principles with those in history education, suggesting cross-domain applicability.
7. **Replication across conflict contexts:** The wartime education findings have potential relevance for other conflict-affected educational settings worldwide [78]. Replication studies in different conflict contexts would test the generalisability of the multi-channel resilience architecture and contribute to the growing body of evidence on technology-enhanced education in emergencies [108, 220].

CONCLUSION

This dissertation addressed the scientific problem of developing a methodology for teaching digital history to future professionals. The following main results were obtained:

1. A PRISMA-guided systematic review [177] of digital history in pedagogical research was conducted, identifying only 6 studies that met rigorous inclusion criteria from an initial pool of 847 records across six databases [123]. This finding establishes the nascent state of digital history pedagogy as a research field and highlights the urgent need for empirical studies. The scarcity of research motivated the examination of gamification as the most extensively studied component of digital history methodology.
2. A comprehensive systematic review of gamification in history education was performed, synthesising evidence from 74 studies [124]. The review revealed: (a) severe definitional ambiguity, with only 14.9% of studies providing an explicit definition of gamification [54]; (b) predominantly positive effects on engagement (70.3% of studies) and learning outcomes (51.4%), consistent with broader gamification meta-analyses [89, 122]; (c) six evidence-based design principles – historical authenticity, narrative integration, progressive difficulty, immediate feedback, social elements, and multiple modalities; (d) four distinct conceptual approaches to gamification (structural, content, full, and game-based learning).
3. A methodological quality assessment of the 74 gamification studies was conducted using the MRQS framework [125, 241], revealing that 83.8% exhibited a high risk of bias. The mean MRQS was 4.31/10 (SD = 1.93, range 0–9). The most critical weaknesses included small sample sizes (66.1% with $N < 50$), lack of control groups (only 31.1% included comparison conditions), absence of operational definitions

(85.1%), and near-universal failure to report effect sizes (90.5%). Risk-of-bias analysis identified participant selection (Domain 1, 85.1% high risk) as the most persistent source of weakness. An LLM validation approach revealed that automated risk-of-bias assessment achieves only slight agreement with human reviewers (Cohen's $\kappa = -0.04$ to 0.21) per the Landis and Koch [131] interpretation scale, while factual data extraction is highly reliable (82–97% agreement), establishing a hybrid model for educational research synthesis.

4. Implementation guidelines for gamification in history education were developed based on the synthesis of systematic reviews and alignment with the European Digital Competence Framework (DigComp 2.0) [232, 238]. The alignment demonstrates that gamified history education simultaneously develops all five DigComp competence areas: information and data literacy, communication and collaboration, digital content creation, safety, and problem solving.
5. A six-month pedagogical experiment was designed and implemented at the Centre for Pre-University Training and Education of Kryvyi Rih State Pedagogical University (October 2024 – April 2025). The course served 23 students and comprised 38 lectures, 30 seminars with 156 Google Forms quizzes, 7 modular controls, and the interactive gamified platform HistoryLikeGame with 23 game modules [127, 249]. This represents one of the most comprehensively documented gamification studies in history education to date [124].
6. A Python-based data analysis pipeline was developed to systematically process all collected data, including chat log parsing, assessment timeline extraction, response analysis, attendance tracking, gamification cataloguing, and figure generation. The pipeline ensures reproducibility and transparency of the analytical process.

7. The findings were validated through cross-domain comparison with a parallel systematic review of gamification in software engineering education (29 studies) [127, 215], which confirmed that the methodological challenges identified in history education are characteristic of gamification research across educational domains, including predominantly positive but uncertain effects (Cohen's $d = 0.23$ for learning outcomes), small sample sizes, and reliance on unvalidated instruments [56].
8. Analysis of the expanded response data (36 seminars spanning the full course, $N=1-27$ per seminar, plus 7 modular controls with $N=7-27$) revealed mean scores ranging from 33.3% to 94.2% across seminars and 63.8% to 84.6% across modular controls. The pre-gamification period (Seminars 1–24) showed a mean of 70.7% across 23 seminars, while the post-gamification period (Seminars 25–39) showed 63.4% across 13 seminars, though this difference is confounded by topic difficulty and assessment format. Over 200 individual data points across 90 named participants were tracked, enabling trajectory analysis. Four distinct performance profiles were identified: consistent high performers (mean 86.9%), variable performers with emerging competence, struggling learners showing improvement through repeated attempts, and declining performers. The modular control data revealed a participation decline from 27 (MC 2) to 7–9 (MCs 5–7), with performance varying by content period.
9. The study provides unique documentation of education under wartime conditions. Attendance analysis identified 33 absence reports, with infrastructure-related causes (internet problems: 24%; power outages: 9%) directly attributable to the armed conflict. The multi-channel course design (Zoom, Viber, Google Drive, Google Forms, HistoryLikeGame) demonstrated resilience through redundancy, ensuring

educational continuity despite infrastructure disruptions.

Scientific novelty. The scientific novelty of the obtained results lies in the fact that:

- *for the first time*, a comprehensive methodology for teaching digital history to future professionals integrating gamification [54] has been developed and theoretically substantiated; a PRISMA-guided systematic review of digital history pedagogy has been conducted [123, 177]; the methodological quality of gamification research in history education has been assessed using MRQS with LLM validation [125]; a gamification-enhanced history education experiment under wartime conditions has been documented [127];
- *improved*: the content and methods of history teaching using digital tools, through the development of an integrated assessment system and implementation guidelines based on DigComp 2.0 [232, 238];
- *further developed*: the theoretical foundations for applying gamification and digital technologies in humanities education [124], and the methodology for systematic reviews with automated quality assessment [125].

Practical significance. The practical results include: (1) the interactive gamified platform HistoryLikeGame; (2) a comprehensive set of 156 quiz forms for Ukrainian history NMT preparation; (3) the data analysis pipeline for processing educational data; (4) implementation guidelines for gamification in history education.

Future research directions.

1. *Expanded data collection*: Collecting complete response data from all seminars and modular controls would enable before-after comparisons and more robust trajectory analysis.
2. *Controlled study*: Implementing a quasi-experimental design with comparison groups

(gamification vs. traditional) would strengthen causal inference [92, 198].

3. *Platform expansion*: Extending HistoryLikeGame to cover all historical periods and adding analytics tracking would provide more comprehensive gamification data.
4. *Larger sample*: Replicating the study with multiple course sections would increase statistical power and generalisability [122, 188, 222].
5. *LLM integration*: Exploring the use of LLMs not only for quality assessment [125, 136] but also for adaptive quiz generation and personalised feedback.
6. *Longitudinal tracking*: Following students through the NMT examination to assess the long-term impact of the preparatory course on examination performance.

Implementation. The results of the dissertation have been implemented in the educational process of the Centre for Pre-University Training and Education of Kryvyi Rih State Pedagogical University (implementation certificate available upon request). The HistoryLikeGame platform continues to be publicly accessible for educational use.

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A Appendices

Appendix A: Course Assessment Timeline

Table A.1 presents the complete timeline of seminar assessments with Google Forms links and the number of quiz blocks per seminar, as extracted from the Viber chat log.

Table A.1: Seminar assessment timeline (complete)

Sem.	Date	Blocks	Naming theme
1	2024-10-20	6	Numbered (Тест 1–6)
2	2024-10-27	5	Numbered (Блок 1–5)
3	2024-11-03	3	Parts (Частина 1–3)
4	2024-11-10	1	Single form
5	2024-11-10	5	Numbered (Блок 1–5)
6	2024-11-17	5	Numbered (Блок 1–5)
7	2024-11-24	4	Numbered (Блок 1–4)
8	2024-12-01	5	Roman numerals (I–V)
8*	2024-12-08	5	Parts (Частина 1–5)
9	2024-11-24	5	Numbered (Блок 1–5)
10	2024-12-01	2	Lecture-based
11	2024-12-15	5	Chapter (Chapter 1–5)
12	2024-12-22	7	Блок 1–5 + Extra 1–2
13	2024-12-29	1	Single form
14	2025-01-05	5	Part 1–5
15	2025-01-12	4	Parts (Частина 1–4)
16	2025-01-19	3	Numbered (Блок 1–3)
17	2025-01-19	5	Greek alphabet (α – ϵ)
18	2025-01-26	3	Numbered (Блок 1–3)
19	2025-02-02	4	Gemstones
20	2025-02-02	5	Greek mythology
21	2025-02-09	5	Greek alphabet (Ukrainian)
22–23	2025-02-16	6	Numbered (Блок 1–6)
24	2025-02-23	4	Numbered (Блок 1–4)
25	2025-03-02	4	Flowers
26	2025-03-09	4	Numbered (Блок 1–4)
27	2025-03-16	4	Trees
28	2025-03-16	5	Number words
29	2025-03-23	4	Flowers (repeated)
30	2025-03-30	4	Minerals

Appendix B: HistoryLikeGame Platform Modules

Table A.2 lists all 23 interactive game modules available on the HistoryLikeGame platform.

Table A.2: Catalogue of HistoryLikeGame modules

Module	Section	Types
Crstm_S_1_1	1 (Ancient/Medieval)	Drag-and-drop, Text input
Crstm_S_1_2	1	Drag-and-drop, Text input
Crstm_S_1_3	1	Drag-and-drop, Text input
Crstm_S_1_3_1	1	Drag-and-drop, Text input
Crstm_S_1_4	1	Drag-and-drop, Text input
Crstm_S_1_5	1	Text input
Crstm_S_1_5_1	1	Drag-and-drop, Text input
Crstm_S_2_1	2 (Early modern)	Text input
Crstm_S_2_2	2	Drag-and-drop, Text input
Crstm_S_2_3	2	Drag-and-drop, Text input
Crstm_S_2_4	2	Drag-and-drop, Text input
Crstm_S_2_5	2	Drag-and-drop, Text input
Crstm_S_3_1	3 (19th–early 20th c.)	Drag-and-drop, Text input
Crstm_S_3_2	3	Drag-and-drop, Text input
Crstm_S_3_3	3	Drag-and-drop, Text input
Crstm_S_3_4	3	Drag-and-drop, Text input
Crstm_S_3_5	3	Drag-and-drop, Text input
Crstm_S_4_1	4 (20th century)	Drag-and-drop, Text input
Crstm_S_4_2	4	Drag-and-drop, Text input
Crstm_S_4_3	4	Drag-and-drop, Text input
Crstm_S_4_4	4	Drag-and-drop, Text input
Crstm_S_4_5	4	Drag-and-drop, Text input
Crstm_S_Omega	Omega (Comprehensive)	Text input

Appendix C: Conferences and Approbation

The results of the dissertation were presented at the following scientific events:

1. **CTE 2025** – 12th Workshop on Cloud Technologies in Education, Kryvyi Rih, Ukraine, 2025. Presentation: “Digital history in educational research: A PRISMA-guided systematic review.”
2. **ICHTML 2025** – 5th International Conference on History, Theory and Methodology of Learning, Kryvyi Rih, Ukraine, 2025. Presentation: “Gamification in history

education: A systematic review and MRQS quality assessment.”

3. **ETQ 2026** – Educational Technology Quarterly Workshop, Kryvyi Rih, Ukraine, 2026. Presentation: “Gamification-enhanced history education under wartime conditions.”

Appendix D: Data Analysis Pipeline

The complete source code for the data analysis pipeline is available in the `thesis/analysis/` directory and consists of the following modules:

1. `parse_chat.py` – Chat log parser (400 messages processed)
2. `extract_assessments.py` – Assessment timeline extractor (156 forms catalogued)
3. `analyze_responses.py` – Google Forms response analyser (12 seminars, 16 named participants)
4. `track_attendance.py` – Attendance and enrollment tracker (23 participants, 33 absences)
5. `catalog_gamification.py` – Gamification platform cataloguer (23 modules)
6. `generate_figures.py` – Publication figure generator
7. `run_pipeline.py` – Master pipeline orchestrator